

CS 425 BLOCKCHAIN TECHNOLOGY

BY

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COURSE CONTENT

- Module 1: Introduction to Blockchain, How Blockchain works, Blockchain vs Bitcoin, Practical applications, public and private key basics, pros and cons of Blockchain, Myths about Bitcoin.
- Module 2: Blockchain Architecture, versions, variants, use cases, Blockchain vs shared Database
- Module3: Introduction to cryptocurrencies, Types, Applications, Concept of Double Spending, Hashing, Mining, Proof of work
- Module 4: Introduction to Merkel tree, Privacy, Payment verification, Resolving Conflicts, Creation of Blocks
- Module 5: Introduction to Ethereum, Advantages and Disadvantages, Ethereum vs Bitcoin
- Module 6: Introduction to Smart contracts, usage, application, working principle, Law and Regulations
- Case Study

SUGGESTED READING

- Lewis, Antony. The basics of bitcoins and blockchains: an introduction to cryptocurrencies and the technology that powers them. Mango Media Inc., 2018.
- Mahankali, Srinivas. Blockchain: The Untold Story: From birth of Internet to future of Blockchain. BPB Publications, 2019.
- Singhal, Bikramaditya, Gautam Dhameja, and Priyansu Sekhar Panda. Beginning Blockchain: A Beginner's Guide to Building Blockchain Solutions. Apress, 2018.
- Wattenhofer, Roger. The science of the blockchain. CreateSpace Independent Publishing Platform, 2016.

COURSE OBJECTIVES

- The objective of this course is to provide conceptual understanding of block chain technology and how it can be used in real time application. The course covers the technological underpinning of block Chain operations in both theoretical and practical implementation of solutions using Ethereum.

COURSE LEARNING OUTCOMES

- Understand block chain technology.
- Understand Cryptocurrency
- Understand Smart contract
- Use Remix IDE
- Develop block chain based solutions and write smart contract using Ethereum Framework
- Deploy Decentralized Application

Module 1 Agenda

- Forms of Money
- Introduction to Blockchain
- How Blockchain works
- Blockchain vs Bitcoin
- Practical applications
- Public and private key basics
- Pros and cons of Blockchain
- Myths about Bitcoin

FORMS OF MONEY

- Barter system
- Commodity money
- Representative money
- Fiat currency

Barter system

- A barter system is an old method of exchange. This system has been used for centuries and long before money was invented. People exchanged services and goods for other services and goods in return.
- The history of bartering dates all the way back to 6000 BC. Introduced by Mesopotamia tribes, bartering was adopted by Phoenicians. Phoenicians bartered goods to those located in various other cities across oceans. Babylonians also developed an improved bartering system. Goods were exchanged for food, tea, weapons, and spices.
- Salt was another popular item exchanged. Salt was so valuable that Roman soldiers' salaries were paid with it. In the Middle Ages, Europeans traveled around the globe to barter crafts and furs in exchange for silks and perfumes. Colonial Americans exchanged musket balls, deer skins, and wheat.

Commodity Money

- With commodity money the physical token that is transacted is itself valuable which have intrinsic or extrinsic value.
- Good forms of commodity money have a stable and known value and are relatively easy to keep and exchange, or 'spend'.
- Commodity money's primary value is in its intrinsic value. This form of money consists of physical items that have a value or utility other than as a form of exchange medium.
- Examples of commodity money include **gold, silver, tea, alcohol, and seashells.**

Representative money

- Representative money is a form of money whose value is derived by being a claim on some underlying item.
- Representative money tokens were similar to bearer bonds, where the person holding a piece of paper was entitled to reclaim the value of the underlying asset.

Fiat currency

- Commodity money was gradually replaced by representative money which in turn has now almost entirely been replaced by ‘fiat money’.
- Fiat money neither has intrinsic value nor is it convertible
- Statements on banknotes often say something along the lines of ‘I promise to pay the bearer on demand the sum of ...’ but you won’t get very far if you go to the issuer of the fiat currency—usually the central bank—and say, ‘Hey, give me some of the underlying asset back for this’. At best you will get a new banknote.

Two main reasons

- They are declared by law as legal tender, meaning that in that legal jurisdiction it must be accepted as valid payment for a debt. Therefore people use it.
- Governments accept only their own fiat for tax payments. This gives fiat currencies a fundamental usefulness, as everyone needs to pay tax.

Early Days

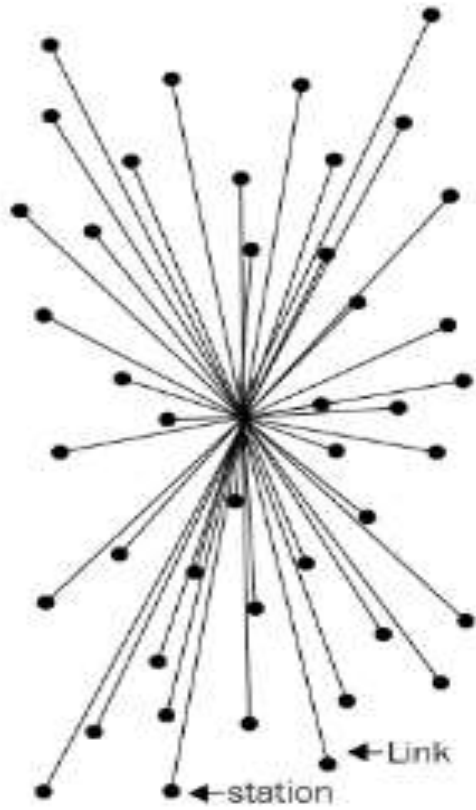
- Bater System to fiat currency – no third party involved
- Difficulty in paying people faraway – banks evolved – time consuming
- Geographically a parted –internet made easy- security & trust
- The technology evolved

The Beginning

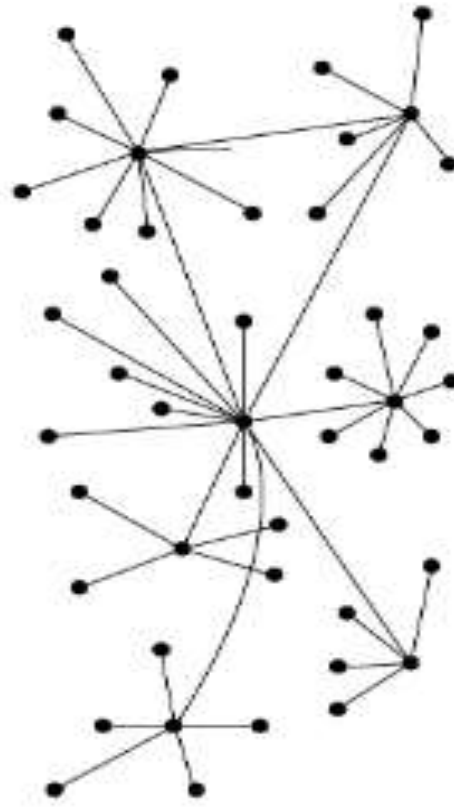
- The conceptual framework behind blockchain was first put forward by a group of researchers in 1991.
- The idea was initially intended for time-stamping digital documents such that backdating them will not be possible thereafter.
- It was again mentioned by Satoshi Nakamoto in his white paper “Bitcoin: A Peer-to-Peer Electronic Cash System”.
- Bitcoin is the first blockchain came into existence in 2019 and bitcoin became popular, and the underlying technology became even more popular.

- So the confusion and lack of clarity among people start from the origin itself; a product and its related terminologies went viral before the technology behind it.
- And when the blockchain displayed its real potential, people were trying to relate it with the bitcoin terminologies; the result was total misconception and confusion.
- But it is the other way; start from blockchain and then try to understand bitcoin.
- To say technology is revolutionary; obviously, it must have a lot of advantage over existing technologies.

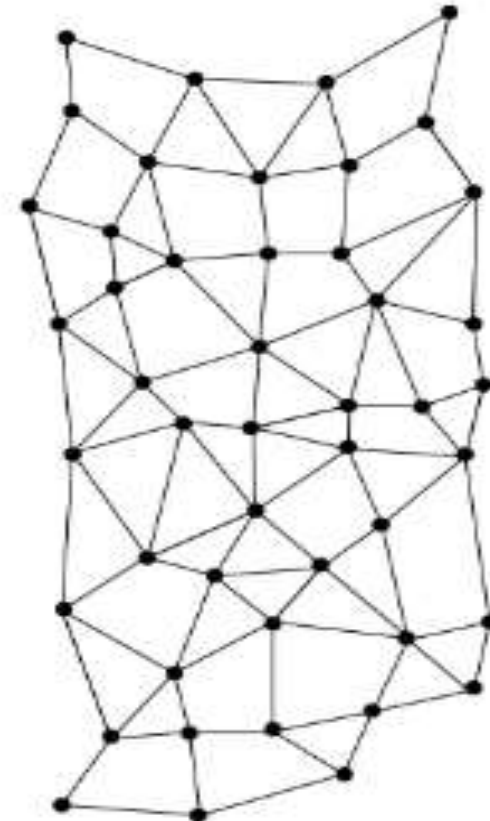
Centralized Vs Decentralized Vs Distributed



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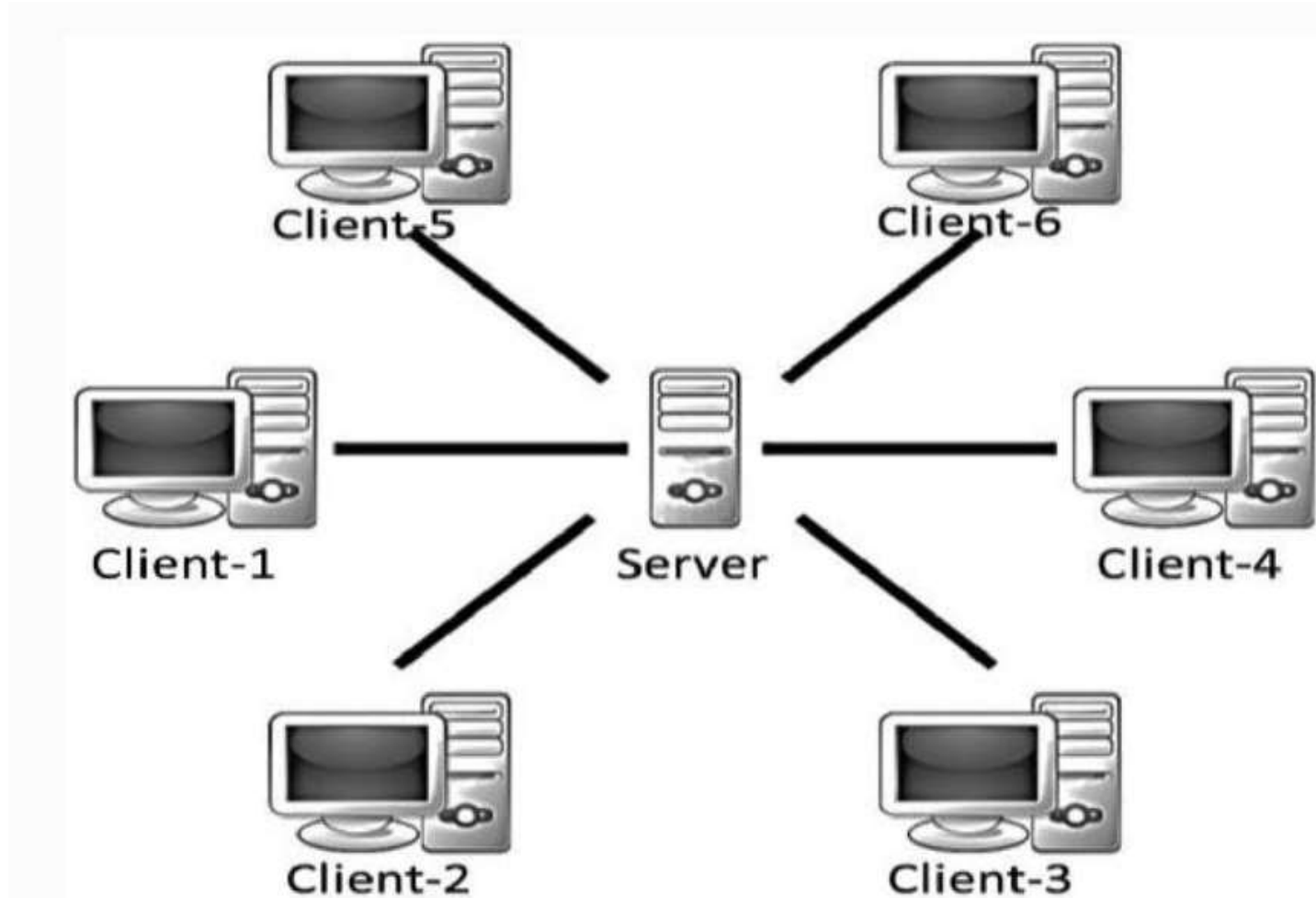


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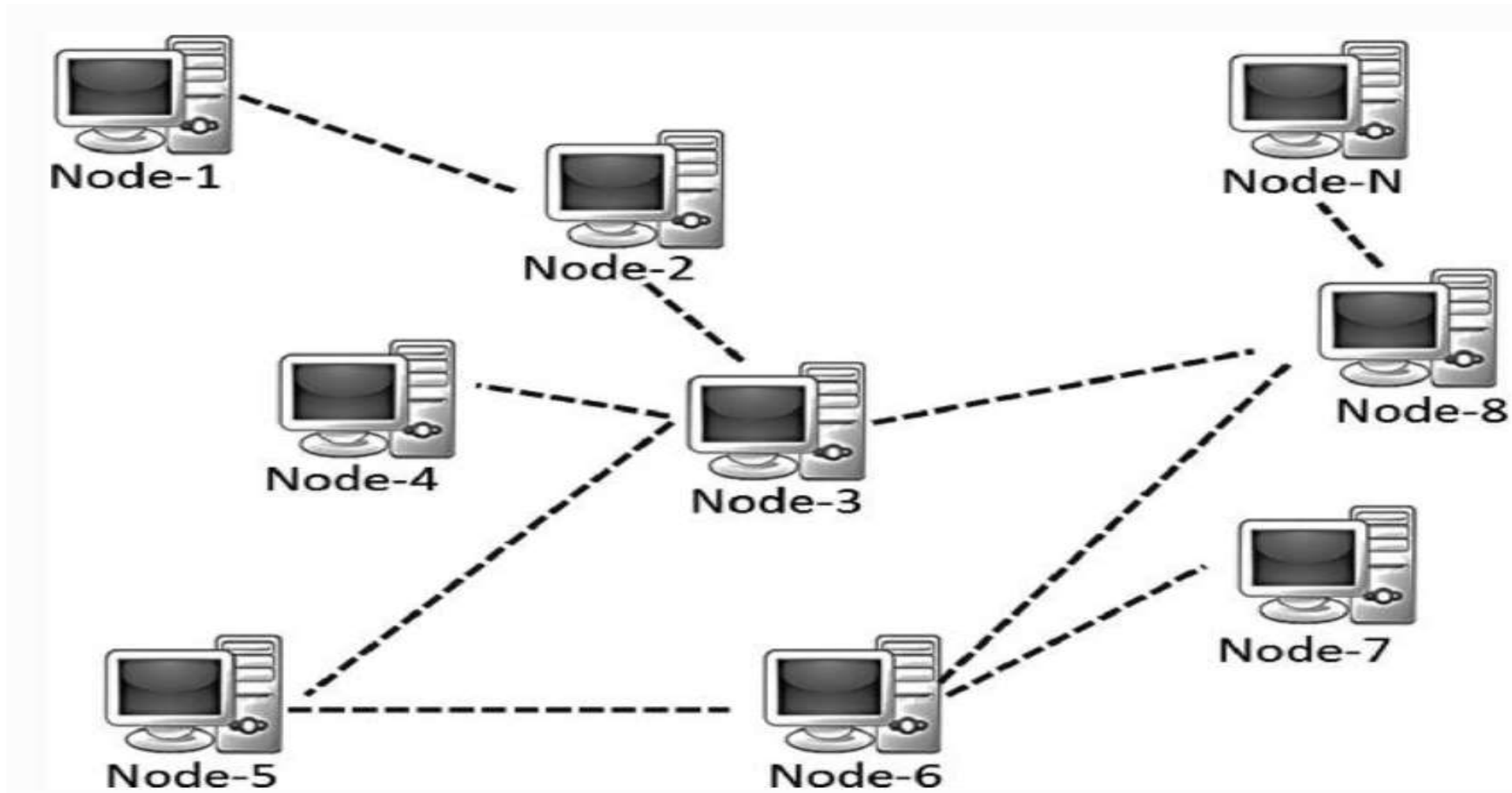


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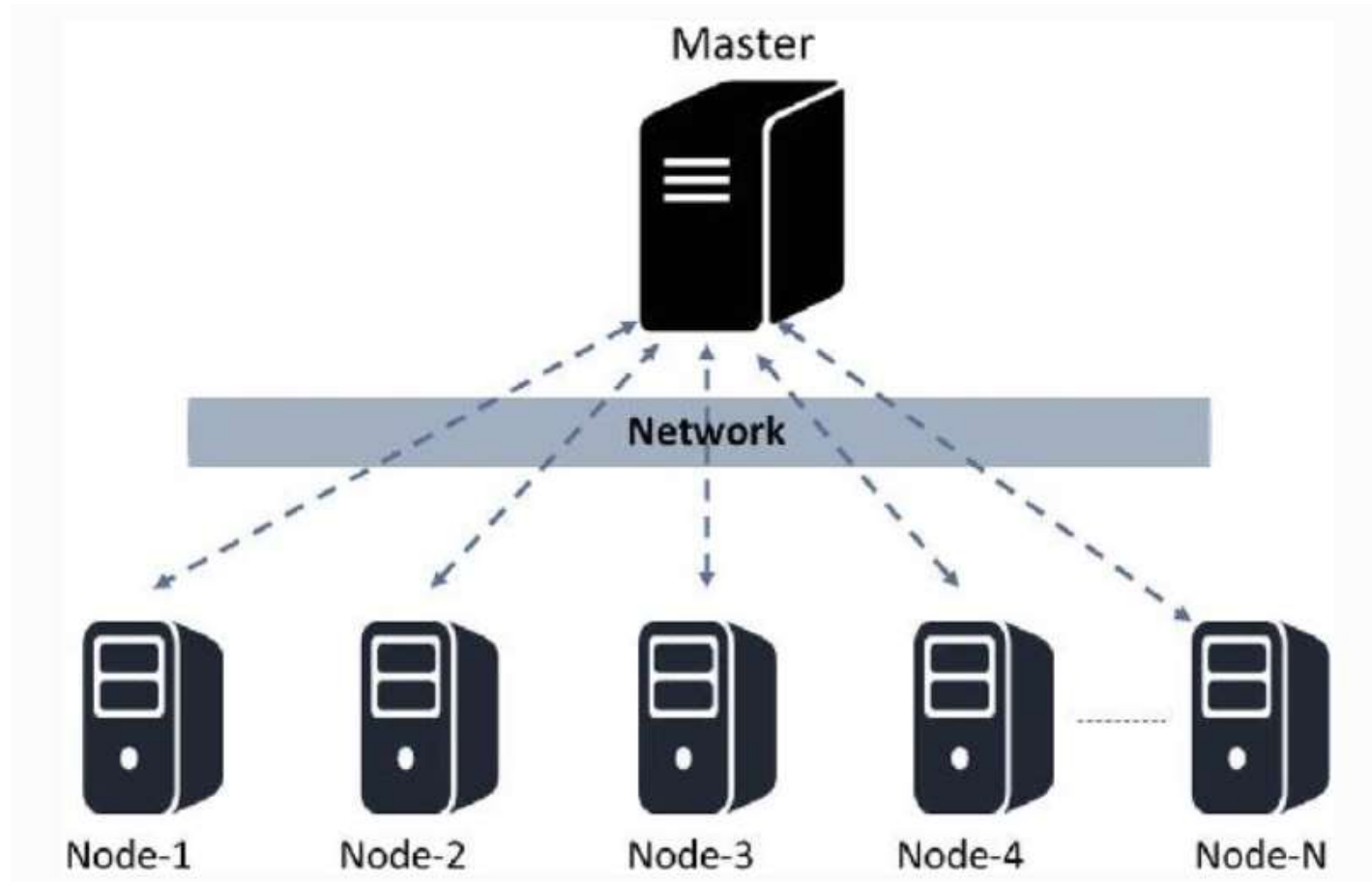
Centralized System



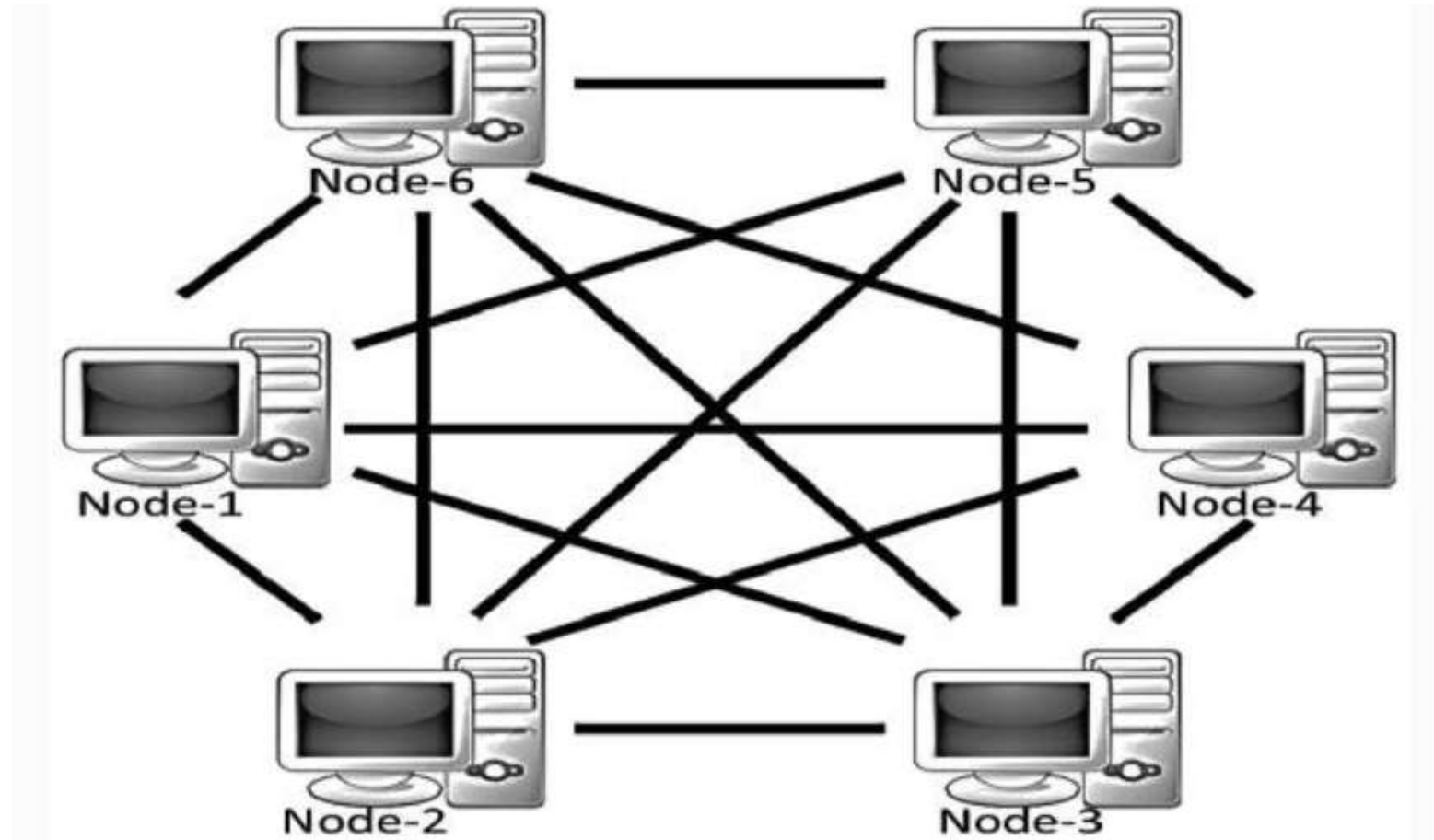
Decentralized Systems



A distributed system with centralized control



A decentralized and peer to peer system



Definitions

- **According to IBM**, blockchain is a shared, distributed ledger that facilitates the process of recording transactions and tracking assets in a network. The asset may be a tangible asset like property, house, vehicle or an intangible asset like digital currency, intellectual property rights, etc.
- ***Layman's definition:*** Blockchain is an ever-growing, secure, shared record keeping system in which each user of the data holds a copy of the records, which can only be updated if all parties involved in a transaction agree to update.

Technical definition

- *A blockchain is “an open, distributed ledger than can record transactions between two parties efficiently and in a verifiable & permanent way”.*
- *Blockchain is a “peer-to-peer, distributed ledger that is cryptographically- secure, append-only, immutable (extremely hard to change), and updateable only via consensus or agreement among peers”.*

- **Peer-to-Peer:** This means that there is no central controller in the network, and all participants talk to each other directly. This property allows for cash transactions to be exchanged directly among the peers without a third-party involvement, such as by a bank.
- **Distributed Ledger:** A ledger is spread across the network among all peers in the network, and each peer holds a copy of the complete ledger.
- **Cryptographically-Secure:** To provide security services which make this ledger secure against tampering and misuse. These services include non-repudiation, data integrity, and data origin authentication.

- **Append-only:** Data can only be added to the blockchain in time-ordered sequential order.
- **Immutable:** This property implies that once data is added to the blockchain, it is almost impossible to change that data and can be considered practically immutable.
- **Consensus:** Critical attribute of a blockchain is updateable only via consensus. It can be done via power of decentralization. No central authority is in control of updating the ledger. To achieve consensus, there are various consensus facilitation algorithms.

Blockchain is a (Features)

- Decentralized
- Distributed
- Secure and Faster
- Transparent and Immutable
- Consensus Mechanisms
- Distributed Ledger
- Smart contract
- Bundle of transactions validated
- Growing list
- Consist of data, timestamp, hash
- Immediately available
- Has a complete history
- User centric –controlled by the contributor
- Interoperability

Why Blockchain

- Resilience
- Time reduction
- Reliability
- Immutable
- Fraud prevention
- Transparency
- Collaboration
- Decentralized

The Term blockchain can be a/an

- **Data Structure:** For a data structure, blockchain refers to data put together into units called blocks. These blocks are connected to one another like a chain, hence the name blockchain.
- **Algorithm:** When used as a name for an algorithm, blockchain refers to a sequence of instructions that negotiates the informational content of many blockchain-data-structures in a purely distributed peer-to-peer system, similar to a democratic voting schema.
- **Suite of technologies:** Blockchain refers to a combination of the blockchain-data-structure, the blockchain-algorithm, as well as cryptographic and security technologies that combined can be used to achieve integrity in purely distributed peer-to-peer systems, regardless of the application goal.
- **Umbrella term for purely distributed peer-to-peer systems with a common application area:** Blockchain can also be used as an umbrella term for purely distributed peer-to-peer systems of ledgers that utilize the blockchain-technology-suite.

Conclude

- The blockchain is a purely distributed peer-to-peer system of ledgers that utilizes a software unit that consist of an algorithm, which negotiates the informational content of ordered and connected block of data together with cryptographic and security technologies in order to achieve and maintain its integrity.

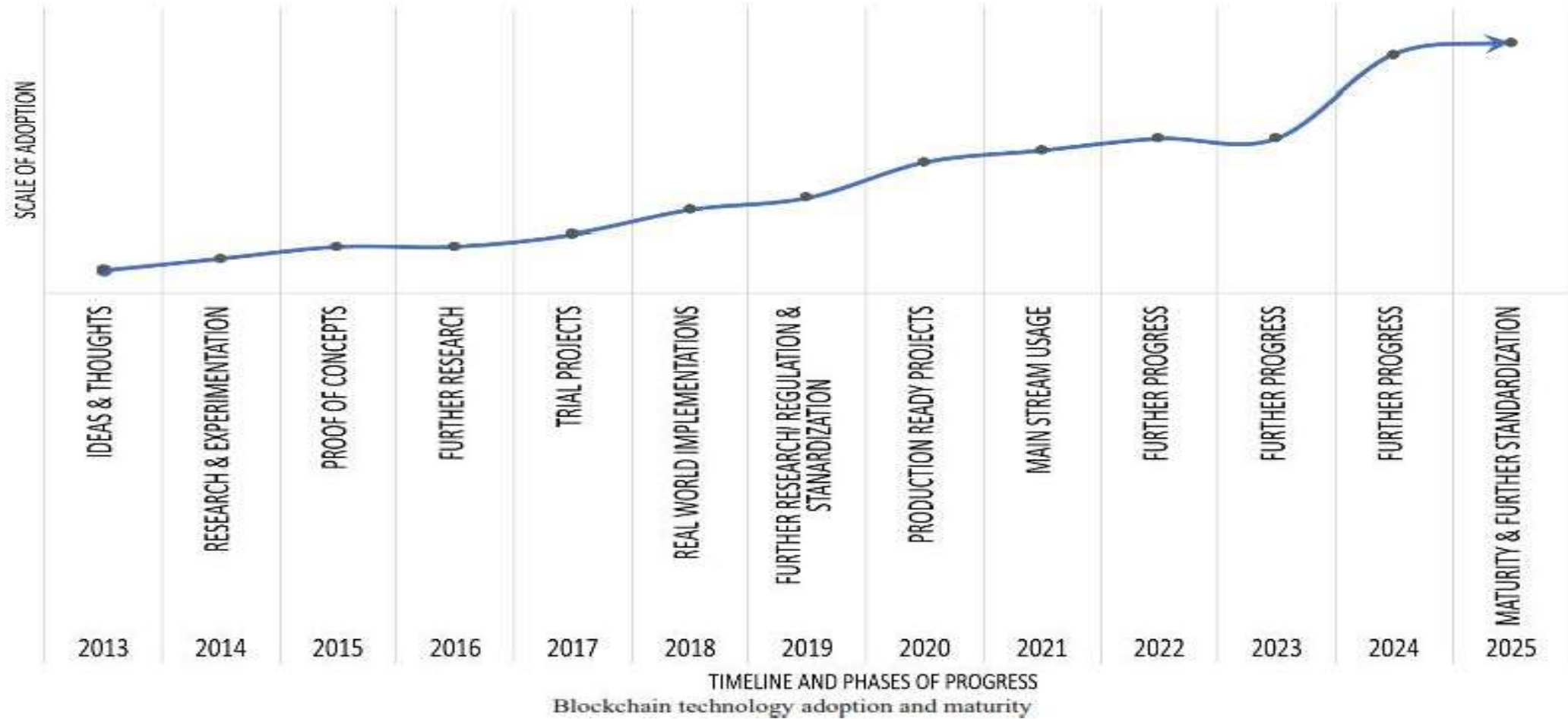
Distributed Ledger

- Distributed ledgers are maintained by a network of nodes, each of which has a copy of the ledger, validates the information, and helps reach a consensus about its accuracy.
- Distributed ledgers have been around for decades but have become more well-known, researched, used, and developed since Bitcoin was introduced.
- Distributed ledgers can be used in nearly every industry where data is collected and used.

Distributed Ledger Technology

- Distributed ledger technology (DLT) is the technological infrastructure and protocols that allow simultaneous access, validation, and record updating across a networked database.
- DLT is the technology blockchains are created from, and the infrastructure allows users to view any changes and who made them, reduces the need to audit data, ensures data is reliable, and only provides access to those that need it.
- DLTs allow information to be stored securely and accurately using cryptography. The data can be accessed using "keys" and cryptographic signatures. Once the information is stored, it can become an immutable database: the rules of the network, written into the coding of the database programming, govern the ledger.

PROGRESS TOWARDS ADAPTION AND MATURITY



CAP THEOREM

- The CAP theorem is a fundamental principle in distributed systems.
- It states that distributed system can only achieve at most two out of the following three guarantees simultaneously:
 - **Consistency:** Every reader receives the most recent write (or an error).
 - **Availability:** Every request (read or write) receives a response, even if some nodes are down.
 - **Partition Tolerance:** The system continues to operate even if network partitions (communication breakdowns between nodes) occur.

BALANCING CAP THEOREM IN BLOCKCHAIN

- **Partition Tolerance and Consistency (CP):** Most blockchains are designed to be partition-tolerant and consistent. They ensure that even if parts of the network are temporarily partitioned, the overall system maintains a consistent state. During network partitions, blockchain systems might experience delays or reduced availability, but they work to ensure that once partitions are resolved, consistency is restored.
- **Availability Vs. Consistency:** Blockchains often make trade-offs between availability and consistency. For instance, during a network partition, some blockchain networks might prioritize maintaining availability and processing new transactions, but this could lead to temporary inconsistencies that are later reconciled.

BALANCING CAP THEOREM IN BLOCKCHAIN

- **Consistency:** Every read receives the most recent write (or an error). In a blockchain context, this means all nodes in the network agree on the current state of the ledger.
- **Availability:** Every request (read or write) receives a response, even if some nodes are down. In the blockchain world, this means that transactions can still be processed and the system remains operational even if some nodes are offline.
- **Partition Tolerance:** The system continues to operate even if network partitions (communication breakdowns between nodes) occur. For blockchains, this means the system can handle network splits and still function correctly.

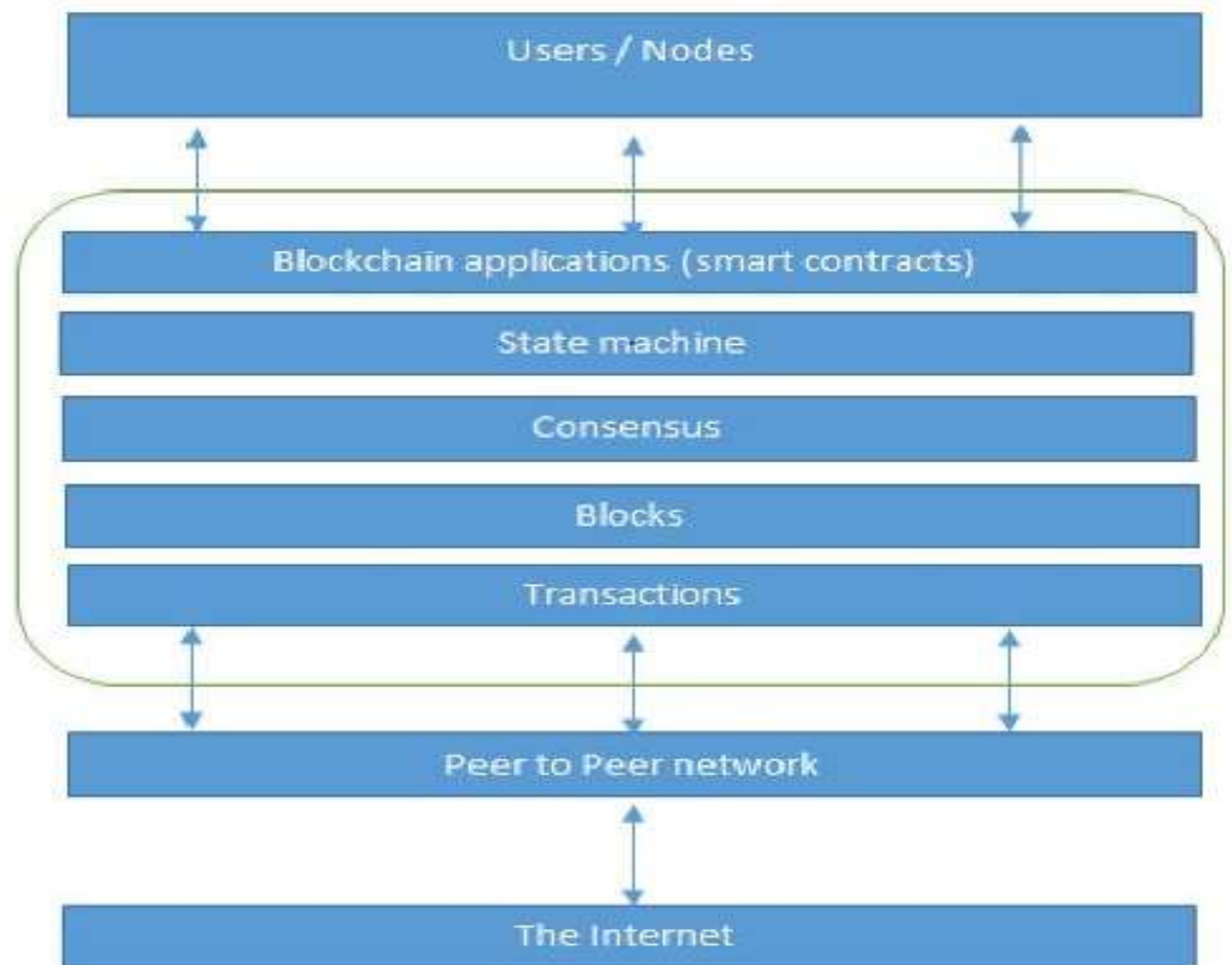
BALANCING CAP THEOREM IN BLOCKCHAIN

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- **Partition Tolerance:** The system continues to operate even if network partitions (communication breakdowns between nodes) occur. For blockchains, this means the system can handle network splits and still function correctly.
- Blockchains are inherently distributed and decentralized, making partition tolerance a crucial aspect. Therefore, many blockchain systems choose to focus on partition tolerance and consistency, sometimes at the expense of immediate availability.

- Consistency and Partition Tolerance (CP): Systems focus on ensuring that all nodes see the same data but may sacrifice availability during network partitions. For instance, some blockchain networks might pause transactions or block confirmations if they detect a network partition.
- Availability and Partition Tolerance (AP): Systems prioritize keeping the service available and operational even if there are network issues, but they might tolerate inconsistencies. This can be seen in blockchains where multiple forks can occur, and different parts of the network might see different versions of the ledger until consensus is reached.
- Consistency and Availability (CA): Achieving both consistency and availability typically requires assuming there will be no network partitions, which is less feasible in a distributed system. Blockchain networks usually aim for this but must adapt when partitions happen.
- Blockchains are inherently distributed and decentralized, making partition tolerance a crucial aspect. Therefore, many blockchain systems choose to focus on partition tolerance and consistency, sometimes at the expense of immediate availability.

BASIC OF BLOCKCHAIN ARCHITECTURE

- Blockchain can be thought of as a layer of a distributed peer-to-peer network running on top of the internet. It is analogous or similar to SMTP, HTTP, or FTP running on top of TCP/IP.



Blockchain Architecture

- At the bottom layer there is the internet, which provides a basic communication layer for any network.
- A peer-to-peer network runs on top of the internet, which hosts another layer of blockchain. This layer contains transactions, blocks, consensus mechanisms, state machines, and blockchain smart contracts. All of these components are shown as a single logical entity in a box, representing blockchain above the peer-to-peer network.
- Finally, at the top, there are users or nodes that connect to the blockchain and perform various operations such as consensus, transaction verification, and processing.

Components of Architecture

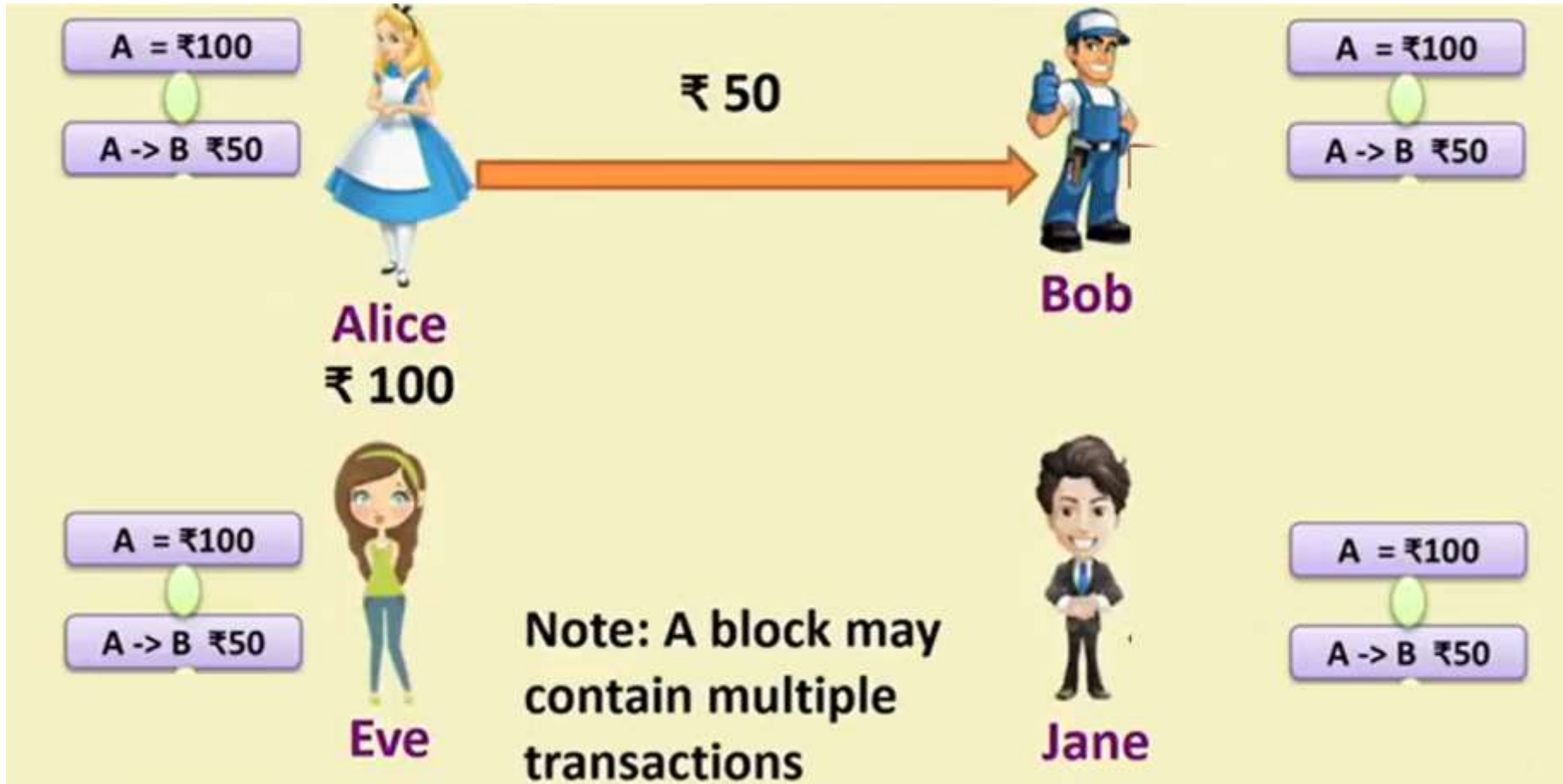
- Transactions
- Blocks
- Consensus mechanisms
- State machines
- Smart contracts

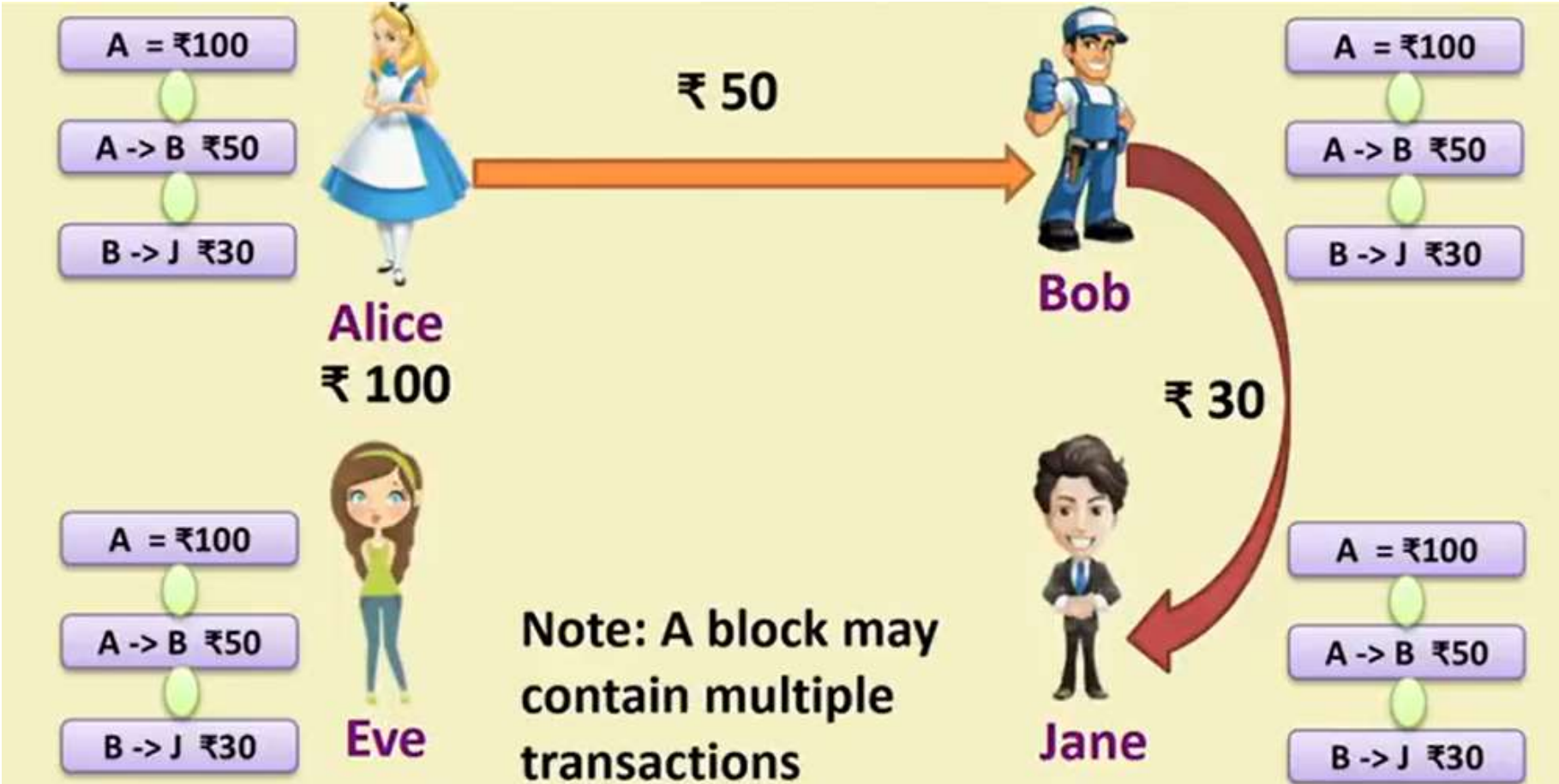
Transaction

- **A transaction** is a record of an event, for example, the event of transferring cash from a sender's account to a beneficiary's account.
- A transaction is the fundamental unit of a blockchain. A transaction represents a transfer of value from one address to another.
- **Address** : Addresses are unique identifiers used in a blockchain transaction to denote senders and recipients. An address is usually a public key or derived from a public key. While addresses can be reused by the same user, addresses themselves are unique.

Example of transaction



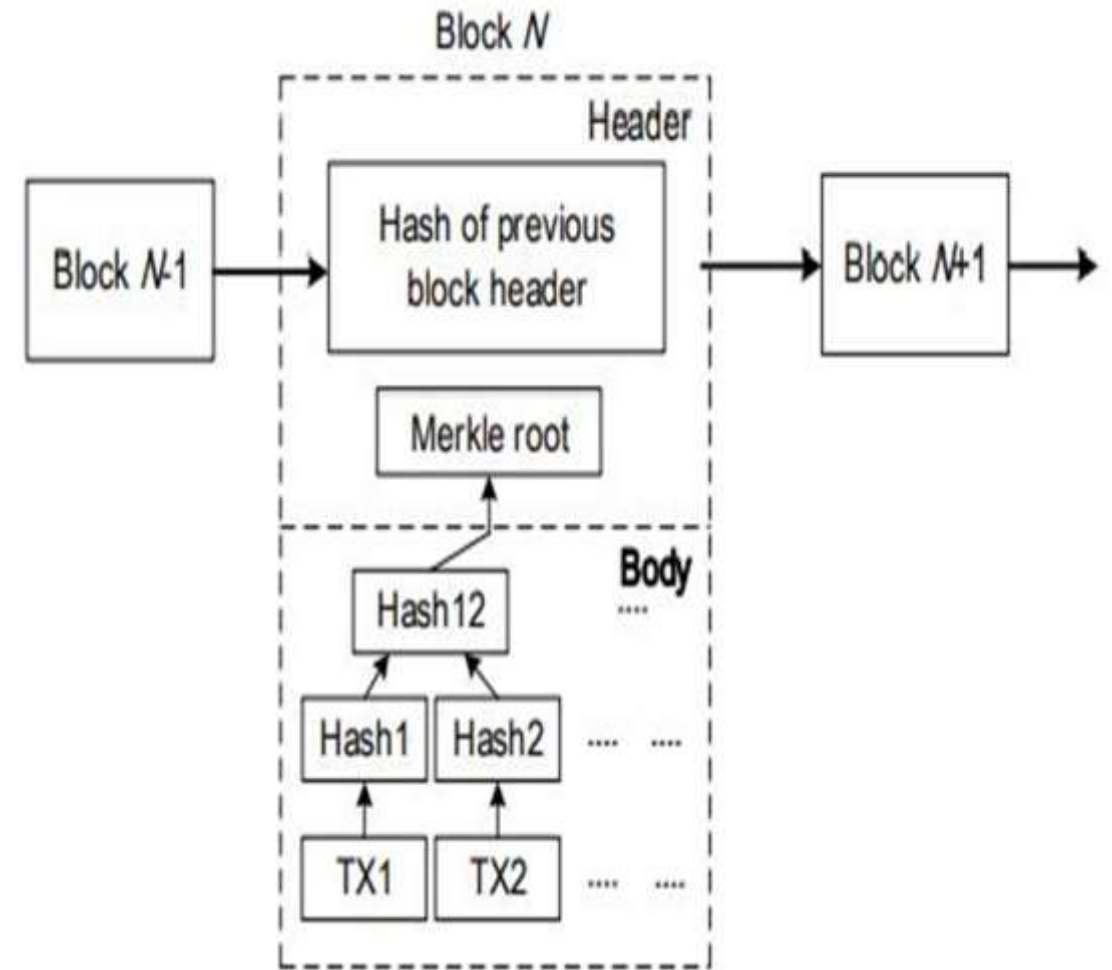
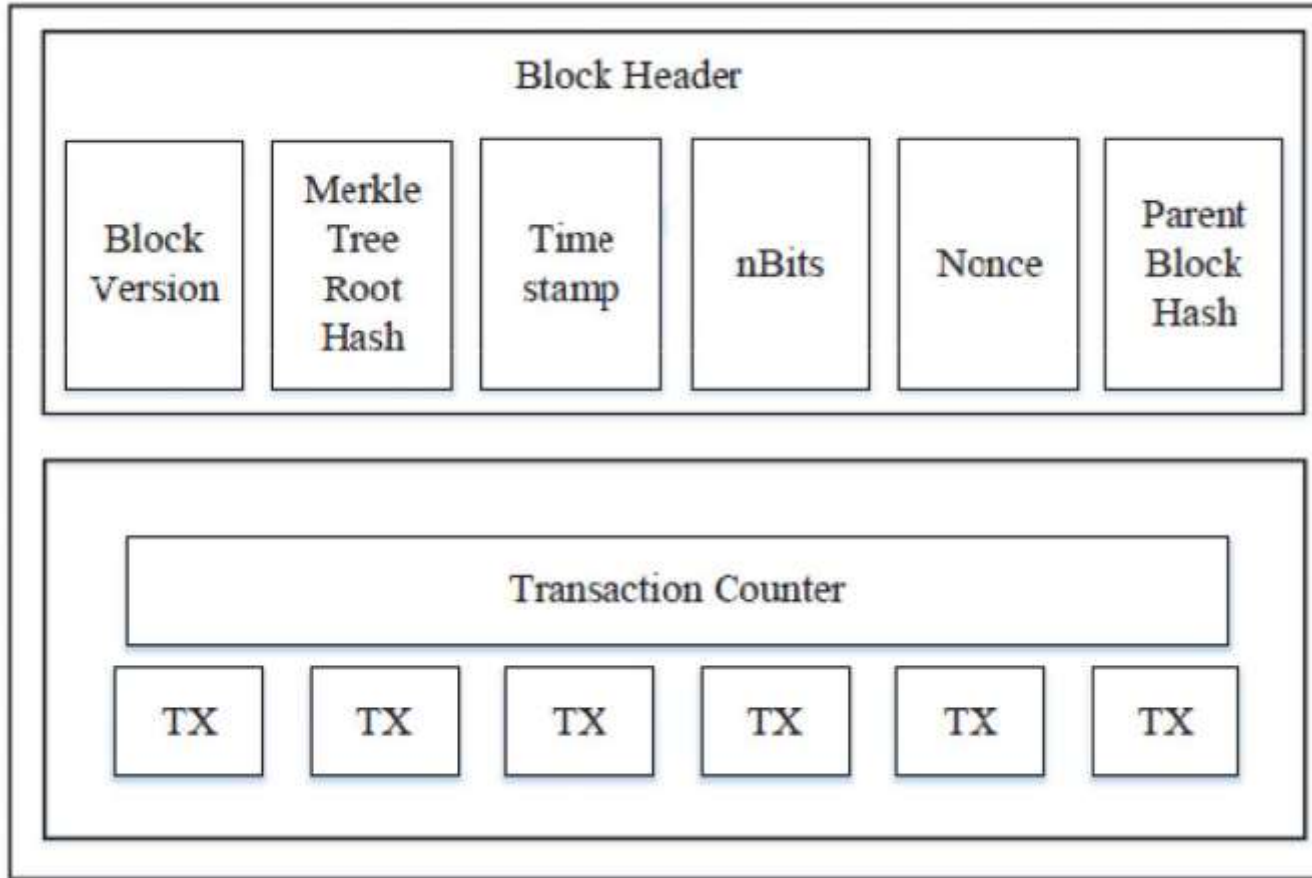




Block

- **A block is merely a selection of transactions bundled together and organized logically.**
- A block is made up of transactions, and its size varies depending on the type and design of the blockchain in use.
- A block is composed of multiple transactions and other elements, such as the previous block hash (hash pointer), timestamp, and nonce.

STRUCTURE OF BLOCK



BLOCK HEADERS ARE SERIALIZED IN 80-BYTE FORMAT

SIZE	FIELD	DESCRIPTION
4 Bytes	Version	A version number to track software/protocol upgrades.
32 bytes	Previous block hash	A reference to the hash of the previous (parent) block in the chain.
32bytes	Merkle root	A hash of the root of the merkle tree of this block's transaction.
4 bytes	Timestamp	The approximate creation time of the block.
4 bytes	nBits Difficulty target	The proof of work algorithm difficulty target for this block
4 bytes	Nounce	A counter used for the proof of work algorithm.

An example header in hex

```
02000000 ..... Block version: 2
```

```
b6ff0b1b1680a2862a30ca44d346d9e8
```

```
910d334beb48ca0c00000000000000000 ... Hash of previous block's header
```

```
9d10aa52ee949386ca9385695f04ede2
```

```
70dda20810decd12bc9b048aaab31471 ... Merkle root
```

```
24d95a54 ..... [Unix time][unix epoch time]: 1415239972
```

```
30c31b18 ..... Target: 0x1bc330 * 256**(0x18-3)
```

```
fe9f0864 ..... Nonce
```

VERSION

The block version number indicates which set of block validation rules to follow:

- Blockchain Version 1.0(cryptocurrency)-It used a public ledger to store the data, for example, Bitcoin.
- Blockchain Version 2.0(smart Contract)- It is called smart contracts which is self-executing programs, for example, Ethereum.
- Blockchain Version 3.0(DAPPS)- It is used to create a decentralized structure, for example, tor Browser.
- Blockchain Version 4.0(Blockchain for Industry)- It is used to create a scalable, affordable blockchain network such that more people could use it.

TIMESTAMP

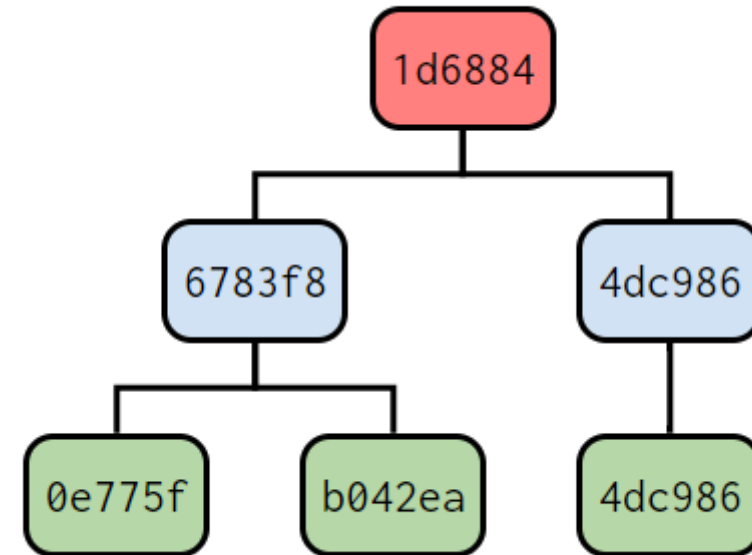
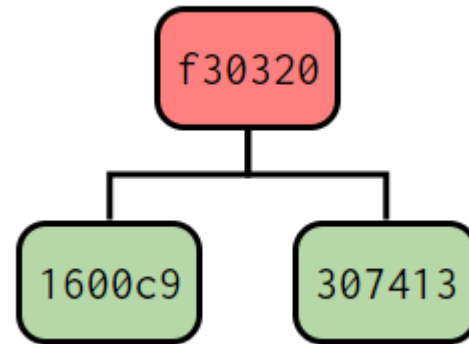
- Timestamp in the blockchain is used as proof that the particular block is used at what instance of a time, also this timestamp is used as a parameter to verify the authenticity of any block.
- The block time is a Unix epoch time when the miner started hashing the header (according to the miner). Must be strictly greater than the median time of the previous 11 blocks. **Full nodes will not accept blocks with headers more than two hours in the future according to their clock.**

MERKLE TREE

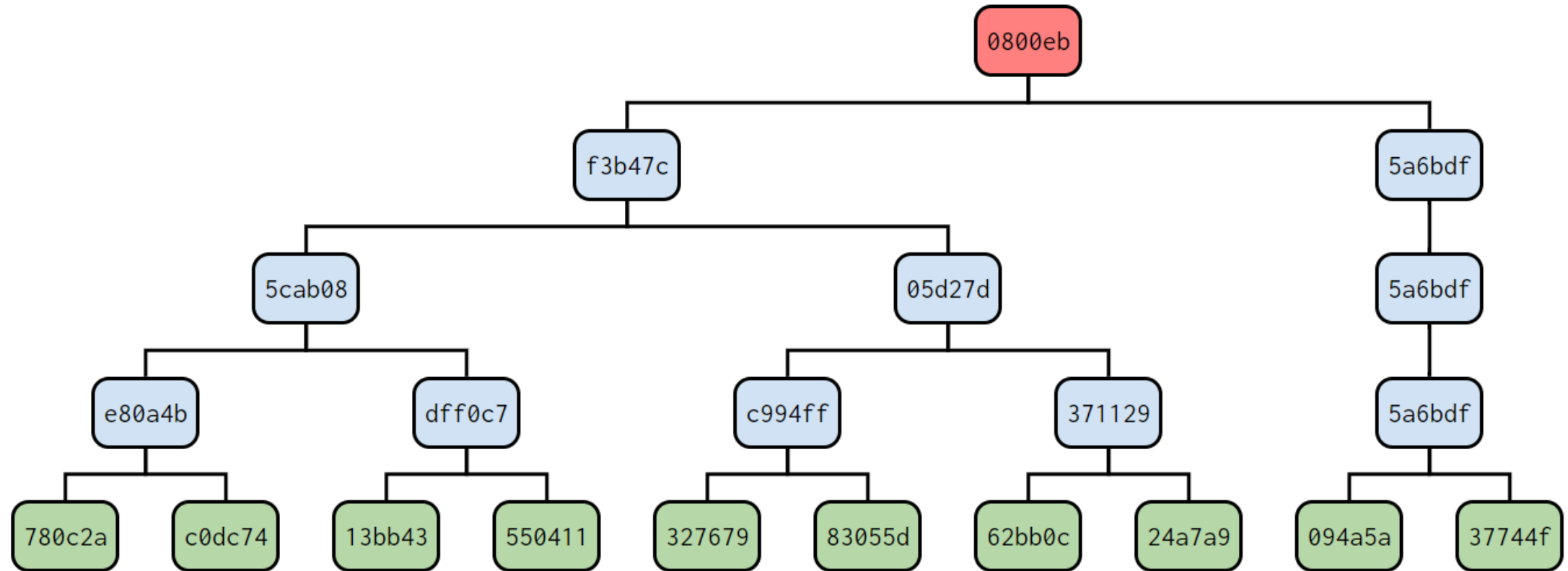
- Merkle tree was introduced by Ralph Merkle in 1979
- It is a form of binary tree that is build using hash function
- Each leaf node corresponds to a piece of data and all the leaf nodes are at the same depth and are as far left as possible
- Every internal node is hash of two children
- It is a data structure used for data verification and synchronization

EXAMPLE OF MERKLE TREE

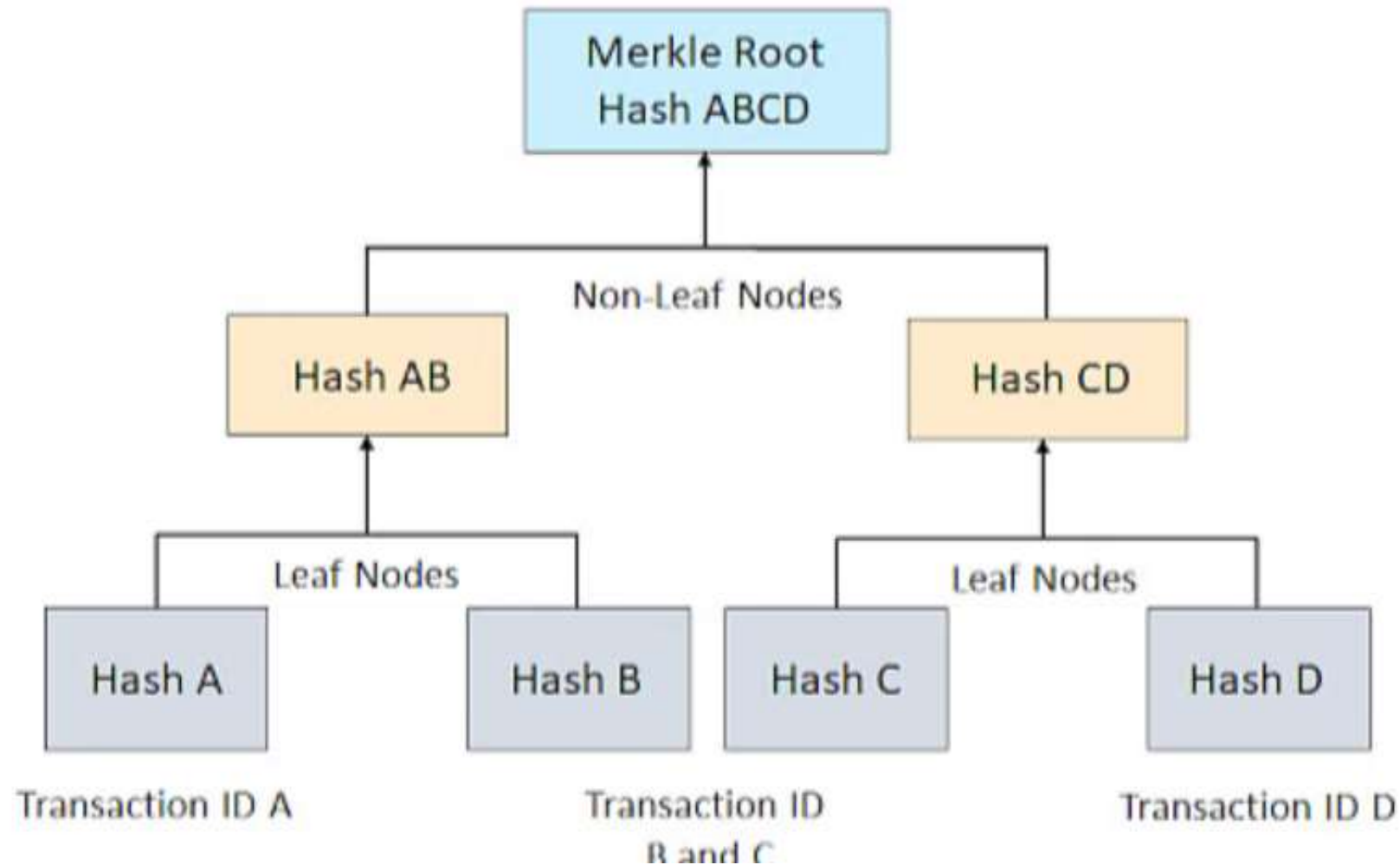
7c64b6



EXAMPLE OF MERKLE TREE



MERKLE ROOT

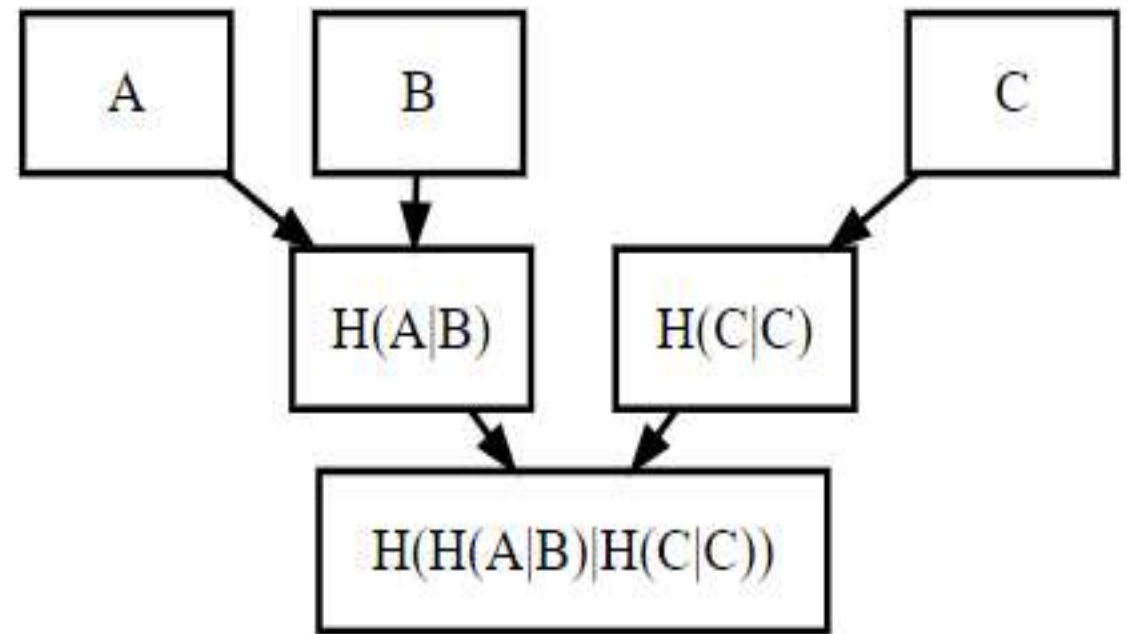


- If a block only has a coinbase transaction, the coinbase TXID is used as the merkle root hash.
- If a block only has a coinbase transaction and one other transaction. the TXIDs of those two transactions are placed in order. concatenated as 64 raw bytes. and then $\text{SHA256}(\text{SHA256}())$ hashed together to form the merkle root.
- If a block has three or more transactions, intermediate merkle tree rows are formed.
- A $\text{SHA256}(\text{SHA256}())$ hash in internal byte order. The merkle root is derived from the hashes of all transactions included in this block, ensuring that none of those transactions can be modified without modifying the header.

Row 1: Transaction hashes (TXIDs)
(A is coinbase; C can spend output from B)

Row 2: Hashes of paired TXIDs

Merkle root

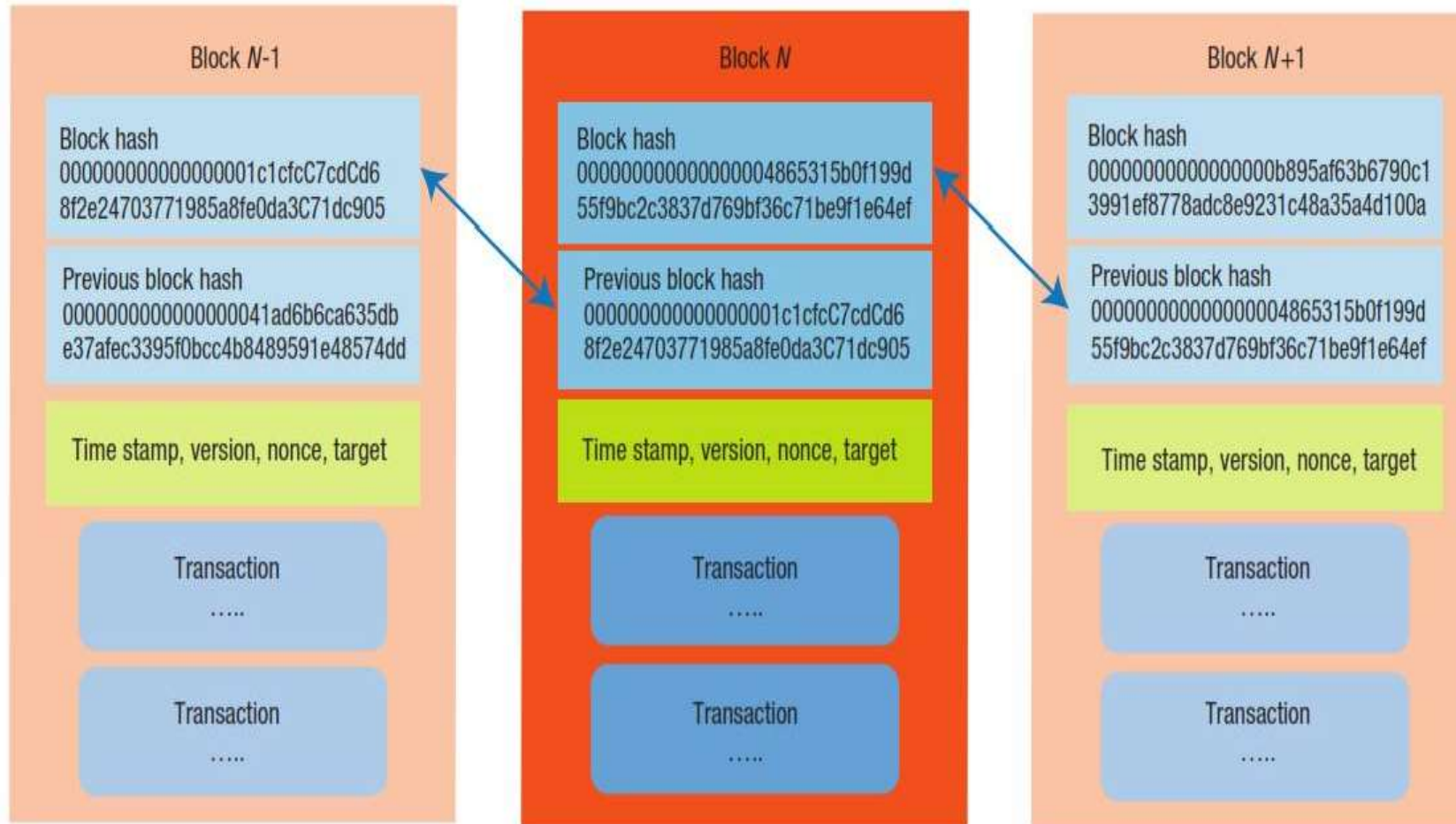


Example Merkle Tree Construction [Hash function $H() = \text{SHA256}(\text{SHA256}())$]

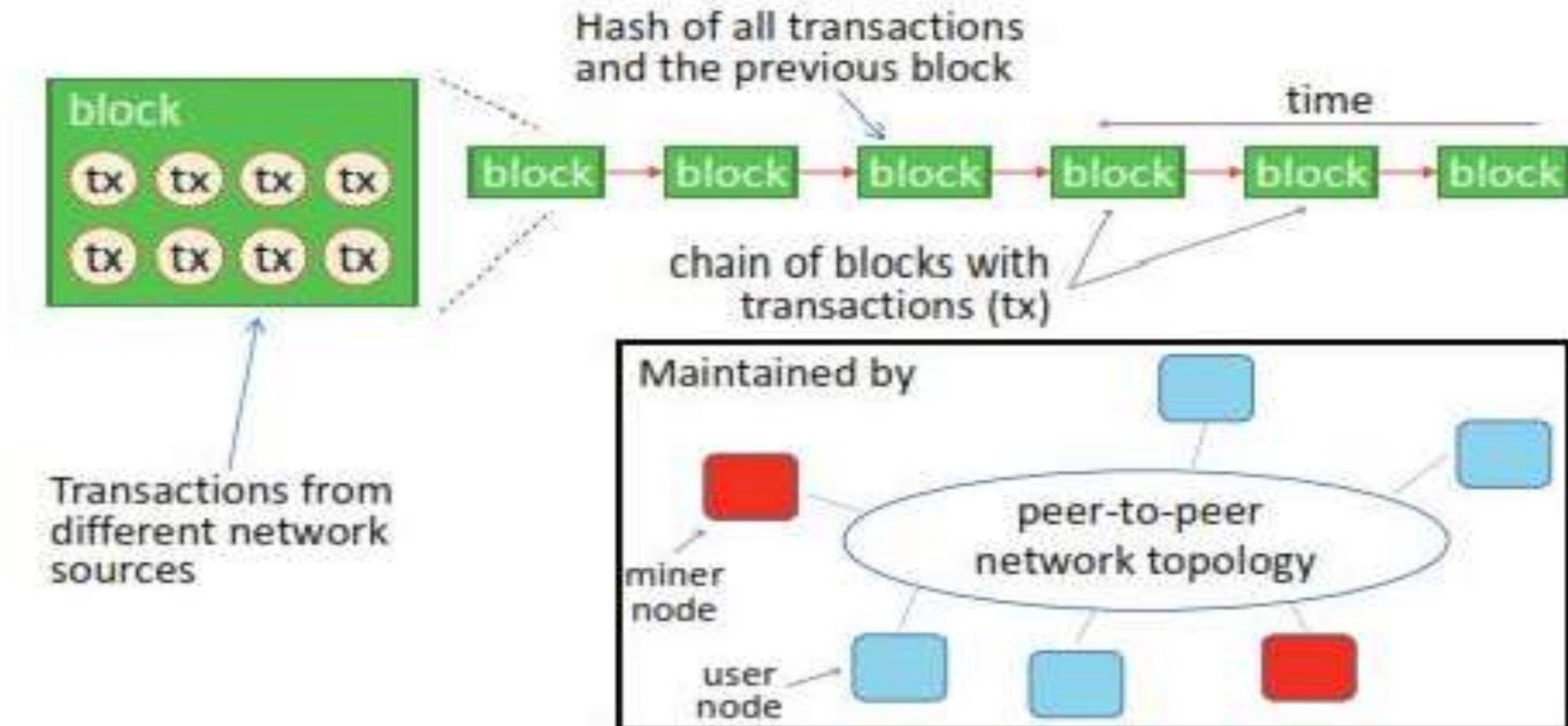
BENEFITS OF MERKLE TREE IN BLOCKCHAIN

- Validate the data's integrity: It can be used to validate the data's integrity effectively.
- Takes little disk space: Compared to other data structures, the Merkle tree takes up very little disk space.
- Tiny information across networks: Merkle trees can be broken down into small pieces of data for verification.
- Efficient Verification: The data format is efficient, and verifying the data's integrity takes only a few moments.

COMPONENTS OF BLOCKCHAIN



OVERVIEW OF BLOCKCHAIN



GENESIS BLOCK

- A **genesis block** is the first block in the blockchain that is hardcoded at the time the blockchain was first started. The structure of a block is also dependent on the type and design of a blockchain.
- **Components of genesis block:**
 - Block header
 - Coinbase transaction
 - Network parameters
 - Embedded message
 - Genesis block hash

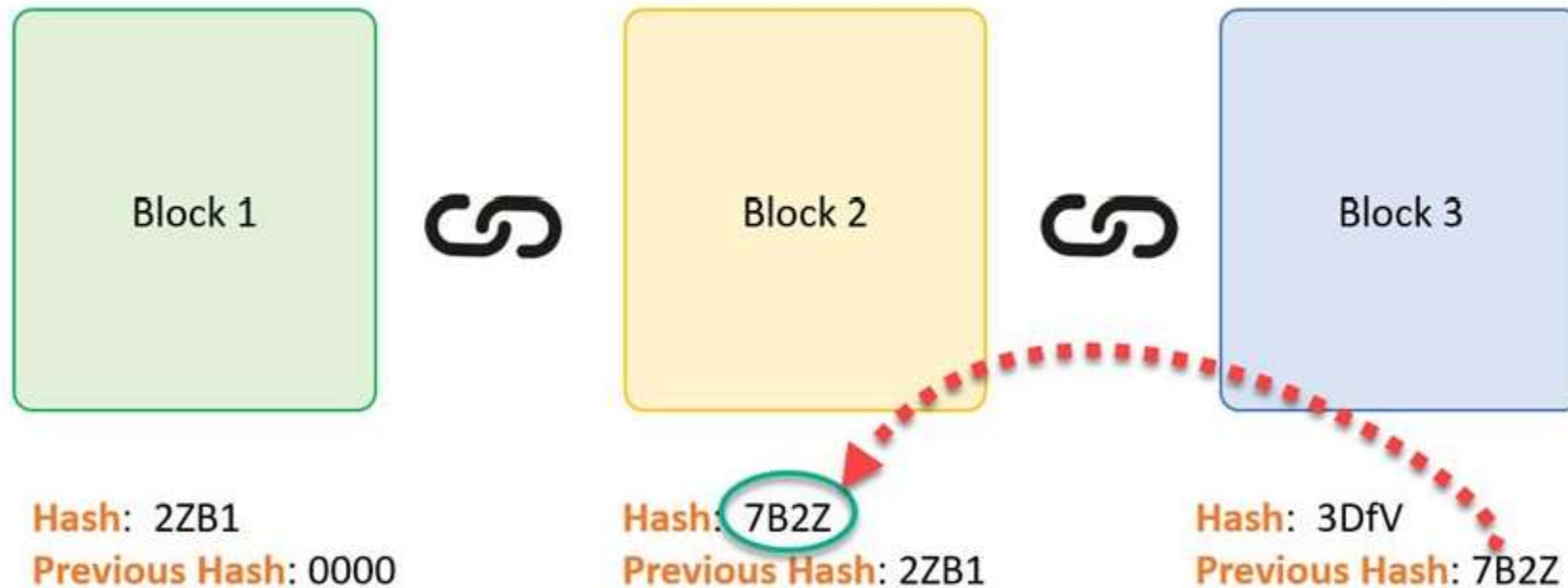
DIFFICULTY TARGET

- It specifies the complexity and the computation power required to mine the network, if we are having a high difficulty target then it implies that we need more a computationally expensive machine to mine it.
- For example, in order to increase the difficulty target algorithms such as SHA-2, SHA-3. RIPEMD, MD5, BLAKE2 is used.

NONCE

- 'number only used once'
- It is a number which blockchain miners are finding and on average, it takes almost 10 times to find out the correct nonce.
- An arbitrary number miners change to modify the header hash in order to produce a hash less than or equal to the target threshold.
- A nonce is a 32-bit number, having the maximum value as 2^{32} total possible value, so the job of the bitcoins miners is to find out the correct integer value which is a random integer between 0 and 2^{32} , so it becomes computationally expensive.

REPRESENTATION OF BLOCKCHAIN



NODE

- **Node:** A node in a blockchain network performs various functions depending on the role that it takes on. A node can propose and validate transactions and perform mining to facilitate consensus and secure the blockchain.

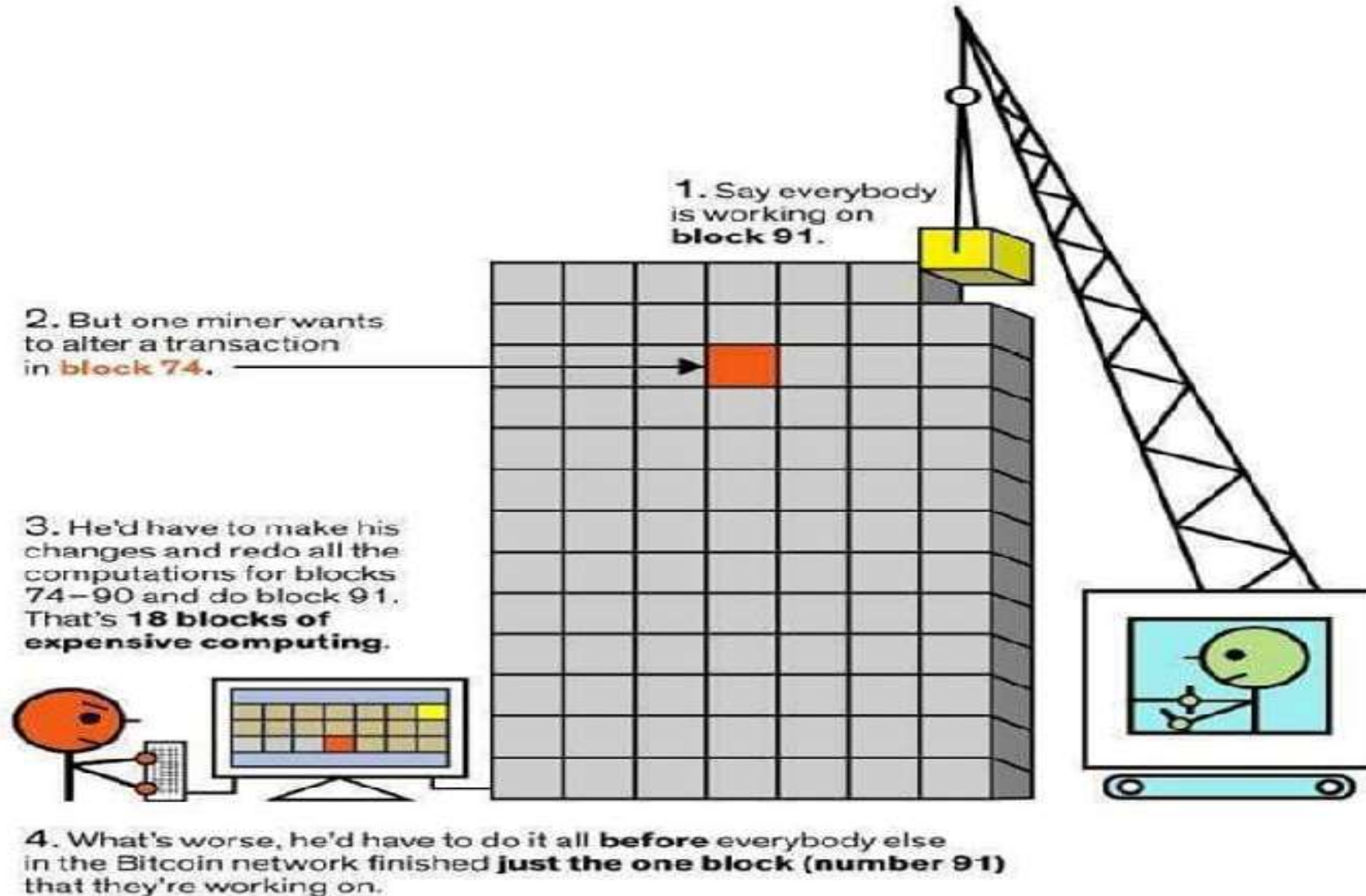
STATE MACHINE

- A blockchain can be viewed as a state transition mechanism whereby a state is modified from its initial form to the next one and eventually to a final form by nodes on the blockchain network as a result of a transaction execution, validation, and finalization process.

SMART CONTRACT

- **Smart contract:** These programs run on top of the blockchain and encapsulate the business logic to be executed when certain conditions are met. These programs are enforceable and automatically executable.
- The smart contract feature is not available on all blockchain platforms, but it is now becoming a very desirable feature due to the flexibility and power that it provides to the blockchain applications.
- Smart contracts have many use cases, including but not limited to identity management, capital markets, trade finance, record management, insurance, and e-governance.

Why blocks can't be change



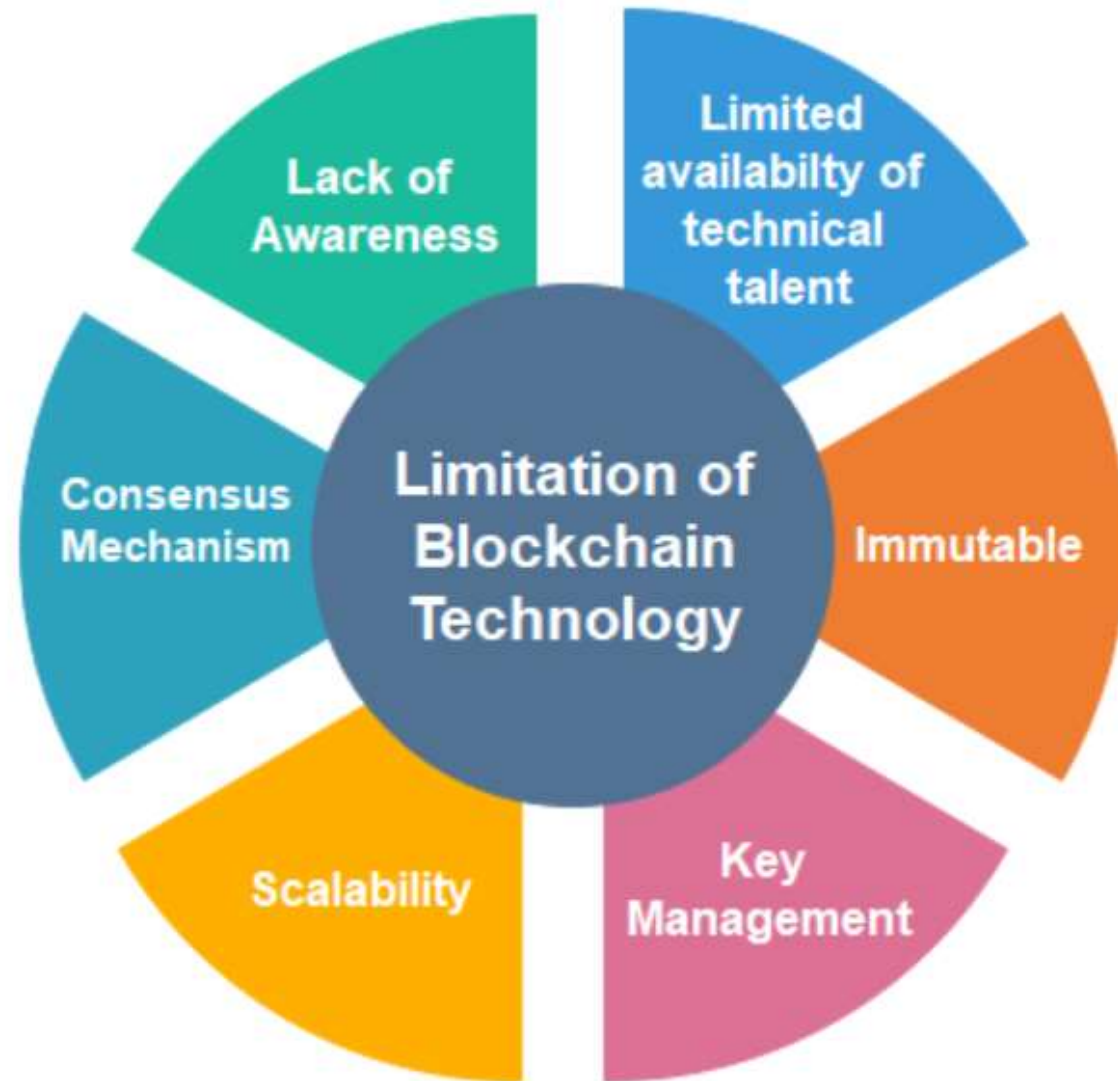
WORKING PRINCIPLE OF BLOCKCHAIN

- **Transaction Initiation:** A node starts a transaction with its private key. A transaction consists of some logic value, relevant rules, source and destination addresses, and other validation information.
- **Transaction Broadcast:** A transaction is propagated (flooded) by using a flooding protocol, called Gossip protocol, to peers that validate the transaction based on preset criteria.
- **Verification:** Once the transaction is validated, it is included in a block, which is then propagated onto the network. At this point, the transaction is considered confirmed.
- **Transaction Pool:** Verified transactions are collected into a pool of pending transaction, awaiting confirmation. This pool is called as mempool.

- **Block Creation:** A miner or a node creates a new block. This involves selecting a set of verified transactions from the pool and grouping them together. The newly-created block now becomes part of the ledger, and the next block links itself cryptographically back to this block. This link is a hash pointer. At this stage, the transaction gets its second confirmation and the block gets its first confirmation.
- **Consensus:** Transactions are then reconfirmed every time a new block is created. Usually, six confirmations in the Bitcoin network are required to consider the transaction final.
- **Chain update:** Once the new block is verified, it is added to the blockchain, extending the chain. All nodes in the network update their copies of the blockchain to include the new block. This ensures that everyone has the same version of the blockchain.
- **Network Consensus:** The blockchain network uses consensus mechanisms (like PoW or Proof of Stake) to agree on the state of the blockchain and ensure all nodes are synchronized. This prevents double-spending and maintains the integrity of the ledger.

ADVANTAGES

- **Decentralization:** There is no need for a trusted third party or intermediary to validate transactions; instead, a consensus mechanism is used to agree on the validity of transactions.
- **Transparency and trust:** Because blockchains are shared and everyone can see what is on the blockchain, this allows the system to be transparent
- **Immutability:** Once the data has been written to the blockchain, it is extremely difficult to change it back.
- **High availability:** As the system is based on thousands of nodes in a peer-to-peer network, and the data is replicated and updated on every node, the system becomes highly available. Even if some nodes leave the network or become inaccessible, the network as a whole continues to work, thus making it highly available. This redundancy results in high availability.
- **Highly secure:** All transactions on a blockchain are cryptographically secured and thus provide network integrity
- **Faster dealing:** Blockchain does not require a lengthy process of verification, reconciliation, and clearance because a single version of agreed-upon data is already available on a shared ledger between financial organizations.
- **Cost saving :** As no trusted third party or clearing house is required in the blockchain model, this can massively eliminate overhead costs in the form of the fees which are paid to such parties.



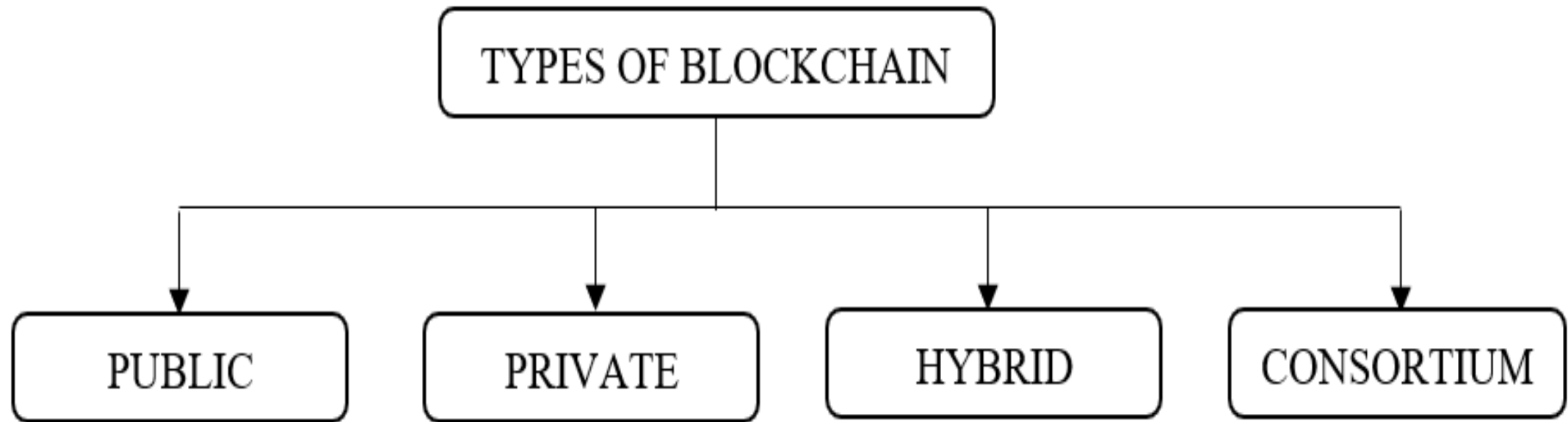
PRACTICAL APPLICATIONS

- Health care industries
- Voting System
- Land Registration
- Agriculture
- Supply Chain Management
- Transportation
- Banking
- Manufacturing
- Food distribution
- Research

TIERS OF BLOCKCHAIN TECHNOLOGY

- Blockchain 1.0: bitcoin
- Blockchain 2.0: financial services
- Blockchain 3.0: other industries
- Blockchain X.0: public

TYPES OF BLOCKCHAIN



Permissionless Blockchain

- It is also known as public blockchains, are available to everyone to participate in the blockchains process that use to validate transactions and data. These are used in the network where high transparency is required.
- Characteristics:
 - Permissionless blockchain has no central authority.
 - The platform is completely open-source.
 - Full transparency of the transaction.
 - Heavy use of tokens.

Private Blockchain

- It is also known as permissioned blockchain as it has restrictions on who can participate in transactions and validations.
- The entities have been chosen in this network by the respective authority and the chain developers while building the blockchain application.
- This blockchain uses are internal to the company so companies will not want it to be accessible by the public.

Characteristics

- Performs better in database management and auditing in other fields.
- No one can run a full node and start mining.
- Everyone cannot create transactions on the chain.
- A user has to be permitted by blockchain authority before he could access the network thus making it Permissioned Network.
- Performance becomes faster as fewer nodes participate.
- It is being able to give service on demand making it more attractive to the user.

Hybrid Blockchain

- A Hybrid blockchain is a blockchain that combines elements of both public and private blockchains. It is designed to reduce the disadvantages of both types of blockchains while maximizing their benefits.
- Similar to public blockchains, where anyone is invited to join and participate in the network, public nodes operate similarly to those in those systems. Private nodes, on the other hand, are in charge of validating and checking transactions and are governed by specific organizations or individuals.
- In a hybrid blockchain, the members can decide who can participate in the blockchain or which transactions are made public.
- A hybrid blockchain essentially has two interfaces: a private blockchain with its own ledgers and a public blockchain that aids in data verification between the private blockchain's ledgers.

Working principle of Hybrid Blockchain

- Once a user is granted access to the hybrid blockchain platform, the user can fully participate in the blockchain's activities.
- The user shares equal rights to do the transactions, view them, or even append or modify transactions.
- The identity of the users is kept a secret from other participants.
- When a user performs a transaction with another user on the network then only his identity is revealed by the party he is dealing with.
- Companies and organizations carry out KYC to ensure that the identification process is done correctly.

Consortium Blockchain

- Consortium blockchains are managed and run by a number of organizations or entities. As a permissioned blockchain, users must be asked to join and have authorization before they can access the network.
- Provide a balance between decentralization and control, making them appropriate for use cases where a number of well-known and trustworthy parties must collaborate on a common platform.
- The upkeep of the blockchain network and transaction verification are divided among the participating groups in a consortium blockchain.
- Instead of being under the control of a singular central authority, the member organizations manage the nodes that verify transactions and keep the blockchain.

Characteristics of consortium blockchain

- Data Privacy: Only a select few individuals have access to the datasets.
- Rapid-Fire Transaction: Controlled user groups can also speed up the consensus-building process.
- No Crime was Committed: The risk of criminal or illegal activity is completely eliminated when a federated blockchain is connected with a small number of well-known members.
- Zero Probability of 51% Act: Attacks from 51% of users could be disastrous for the blockchain network. A private blockchain is more vulnerable to this problem because multiple groups' members can work together to override or even perform reverse transactions.

BITCOIN

- Bitcoin is a completely decentralized, peer-to-peer, permissionless cryptocurrency put forth in 2009.
- Currently the most successful digital currency in the world with billions of dollars invested in it.
- Bitcoin is a decentralized cryptocurrency that is not limited to any nation and is a global currency. It is decentralized in every aspect—technical, logical, as well as political.
- As and when the transactions are validated, new Bitcoins get mined and a maximum of 21 million Bitcoins can ever be produced. Approximately, to reach 21 million Bitcoins, it would take until the year 2140.
- The Bitcoins can have any value with eight decimal places of precision. So, the smallest value in Bitcoin is 0.00000001 BTC, which is called 1 Satoshi.

- The name of the Bitcoin inventor Satoshi Nakamoto is believed to be a pseudonym, as the true identity of Bitcoin inventor is unknown.
- Key idea of bitcoin was a peer-to-peer electronic cash that does not need an intermediary bank to transfer payments between peers.

Bitcoin can be defined as

- Platform / network
 - Digital cryptocurrency
 - Protocol
-
- It is a combination of peer-to-peer network, protocols, software that facilitate the creation and usage of the digital currency named bitcoin. Nodes in this peer-to-peer network talk to each other using the Bitcoin protocol.

Bitcoin as network

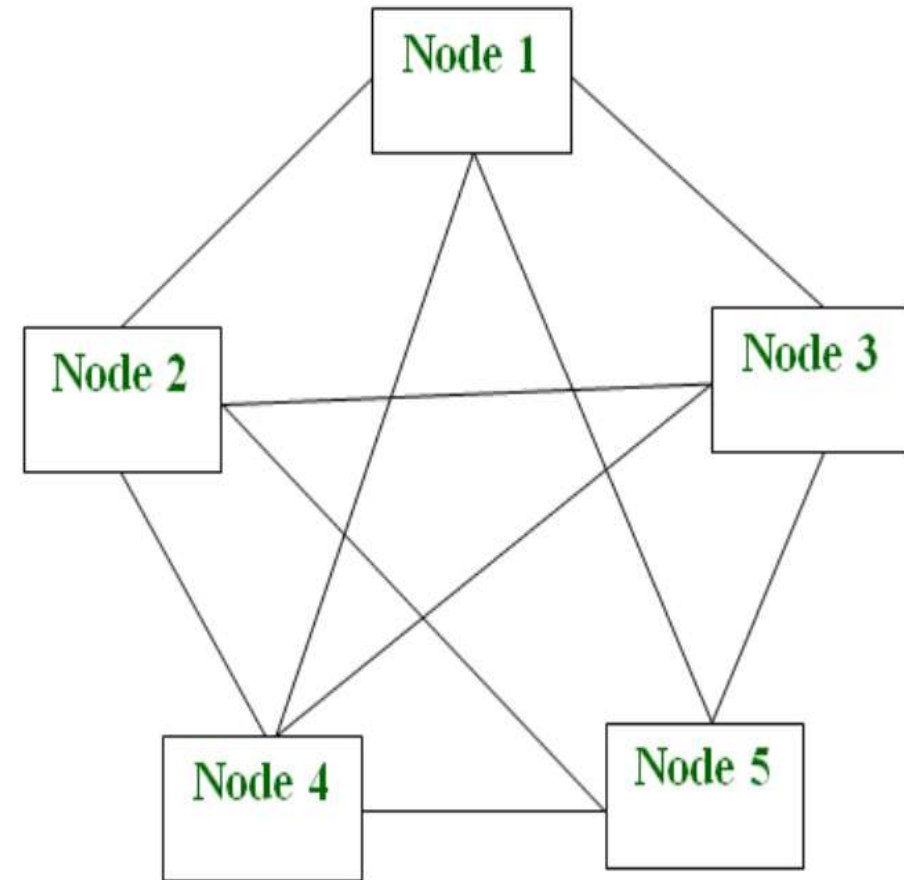
- The main components of bitcoin network are :
 - Digital keys
 - Addresses
 - Transactions
 - Blockchain
 - Miners
 - The Bitcoin network
 - Wallets (client software)

BITCOIN NETWORK

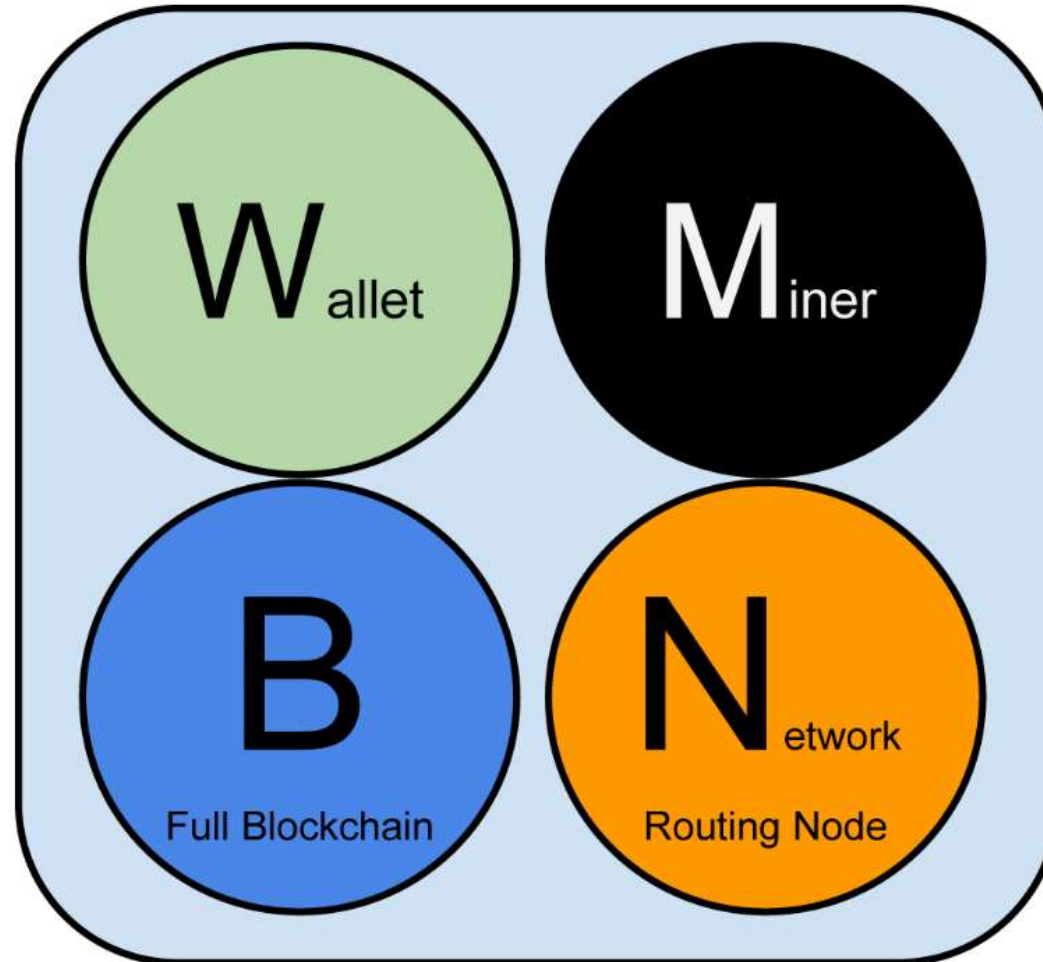
- The Bitcoin network is a peer-to-peer network where nodes exchange transactions and blocks.
- Although nodes in the bitcoin P2P network are equal, they may take on different roles depending on the functionality they are supporting.
- A bitcoin node is a collection of functions: routing, the blockchain database, mining, and wallet services.
- Bitcoin nodes are computers or devices that participate in the Bitcoin network by maintaining a copy of the entire blockchain, verifying transactions, and relaying information to other nodes. They form the backbone of the decentralized Bitcoin network, playing a crucial role in its security, integrity, and functionality.
- Each node in the network operates using Bitcoin software, which enables it to connect to other nodes through the peer-to-peer protocol.
- There are different types of nodes on the network: Full node and SPV node.
- Mining node: they are solo miner nodes that can perform mining, store full blockchain and act a bitcoin network routing node.

P2P NETWORK ARCHITECTURE

- Each computer in the network has the same set of responsibilities and capabilities.
- Each device in the network serves as both a client and server.
- In the peer-to-peer network the file transfer load is distributed among the peer computers.



BITCOIN NETWORK NODE WITH ALL FOUR FUNCTIONS



Types of Nodes

- Full Node
- Light weight / SPV Nodes
- Mining Node
- Listing Node
- Prune Node

FULL NODE

- Full nodes are also referred to as **archival full node**. Archival full nodes are servers that store a copy of the blockchain in its entirety, with every single transaction recorded in their databases.
- As the name implies, they are implementations of Bitcoin core clients performing the wallet, miner, full blockchain storage, and network routing functions. However, it is not necessary to perform all these functions.
- Full node can do almost every permissible activity in the Bitcoin system such as mining transaction, broadcasting transaction and provide wallet service.
- The primary task of these nodes involves validating blocks and maintaining consensus.

- To become a full node one have to download the entire blockchain database that contain the entire transaction that take place till present.
- The node must stay permanently connected to the Bitcoin network and hear all transaction taking place.
- It is important that you have a good network connection, good storage (at least 200 GB) and atleast 2 GB RAM dedicated to it.

Light / SPV Nodes

- Lightweight nodes or “light nodes” or Simplified Payment Verification.
- They don't hold full copies of the blockchain instead they rely on full nodes to verify transaction.
- Light nodes are more accessible for casual users, that only downloads blockheaders, saving users storage space , download time and computational resources.
- However, they sacrifice some level of security and trustlessness compared to full nodes, as they rely on trusted full nodes for transaction verification.

MINING NODES

- Mining nodes, also known as miners, perform the resource-intensive process of mining new bitcoins.
- Miners are either full or light nodes that try to prove they've completed the work required to create a new block.
- This is where the term "proof-of-work" originates. These nodes compete to solve complex mathematical puzzles, known as proof-of-work, to add new blocks to the blockchain.
- To perform this task, miners must either be an archival full node themselves, or get data from other nodes to learn the current status of the blockchain so they can determine how to work on finding the next block.

PRUNED FULL NODES

- Pruned nodes are a variation of full nodes that store a pruned version of the blockchain. They discard older blocks while retaining only a subset of the blockchain data.
- A pruned full node is one that uses less hard drive space by “pruning” older blocks in the chain.
- This type of node will first have to download a copy of all transactions in the blockchain. Then it will start deleting blocks beginning with the oldest and continue until the node only holds the most recent transactions up to a predetermined limit.
- If a node operator were to set the size limit to 300 MB, then a pruned node would only contain the most recent 300 MB worth of transactions.

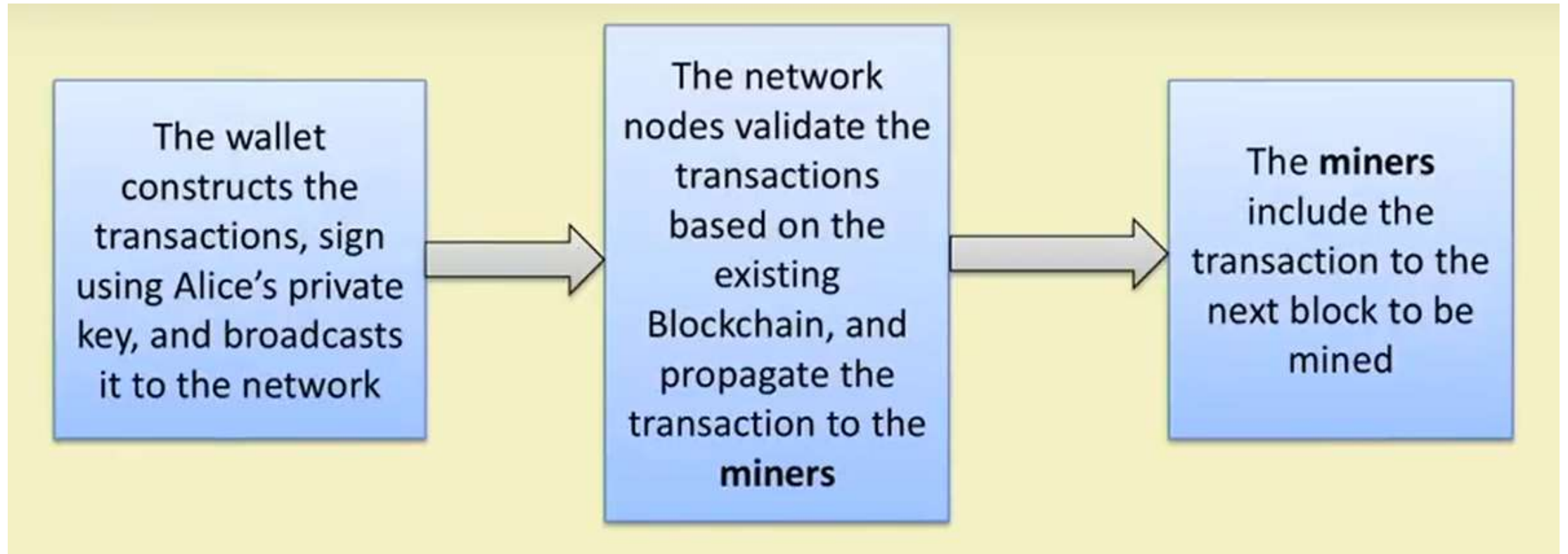
LISTENING NODES

- Listening nodes act as network participants, but they do not store the blockchain. Instead, they focus on relaying information between nodes. Listening nodes receive transactions and blocks, forwarding them to other nodes in the network.
- While they do not validate transactions themselves, they help propagate information and facilitate network communication.

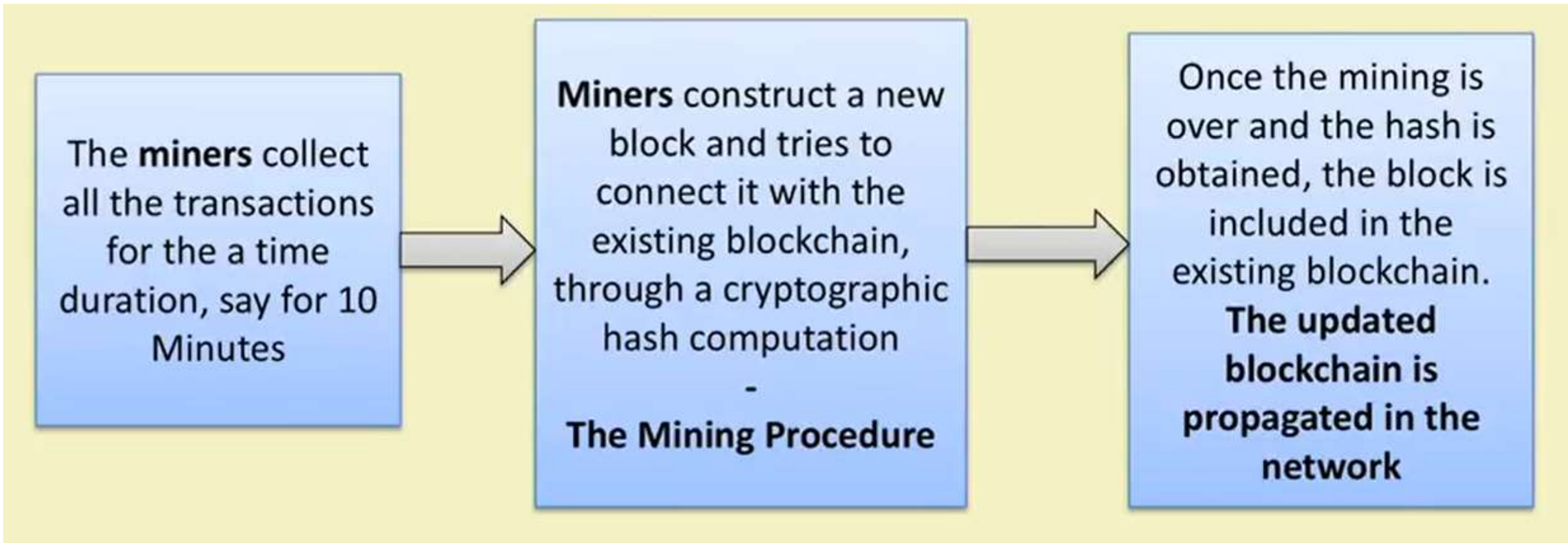
The bitcoin transaction life cycle-the receiver



The bitcoin transaction life cycle



The bitcoin transaction life cycle



The bitcoin transaction life cycle-the receiver

Bob opens his Bitcoin Wallet and refreshes, the blockchain gets updated

The transaction reflects at Bob's wallet



Bitcoin

UGX

rate

708.41

balance

337952.50

USD

rate

0.24

(default) balance

112.44

UYU

rate

6.17

balance

2944.04

UZS

rate

593.90

balance

283322.42

VEF

rate

1.50

balance

713.64

VND

rate

5112.73

balance

2439059.32

VUV

rate

25.04

balance

REQUEST COINS

SEND COINS

SCAN

FILTER

mBTC477.06

+ USD 112.44

Apr 30

1CQh RcTg c4KA MFFB xDdY vYNA r fAJ...

- 4.20

Apr 22

1Nab NwQ3 9Hnr mDYF Nwvw dQdg mmm...

- 21.29

April 21, 15:18

18CK 5k1g ajRK
KSC7 yVST XT9L
Uzbh eh1X Y4

+ 6.26

Apr 17

18CK 5k1g ajRK KSC7 yVST XT9L Uzbh...

+ 13.09

PAYMENT TRANSACTION IN THE BITCOIN NETWORK

- Transaction starts with a sender signing the transaction with their private key
- Transaction is serialized so that it can be transmitted over the network
- Transaction is broadcasted to the network
- Miners listening for the transactions picks up the transaction
- Transaction are verified for their validity by the miners
- Transaction are added to the candidate/proposed block for mining
- Once mined, the result is broadcasted to all nodes on the Bitcoin network

BITCOIN ADDRESS

- A bitcoin address is a string of digits and characters that can be shared with anyone who wants to send you money.
- Addresses produced from public keys consist of a string of numbers and letters, beginning with the digit “1”.

1J7mdg5rbQyUHENYdx39WVWK7fsLpEoXZy

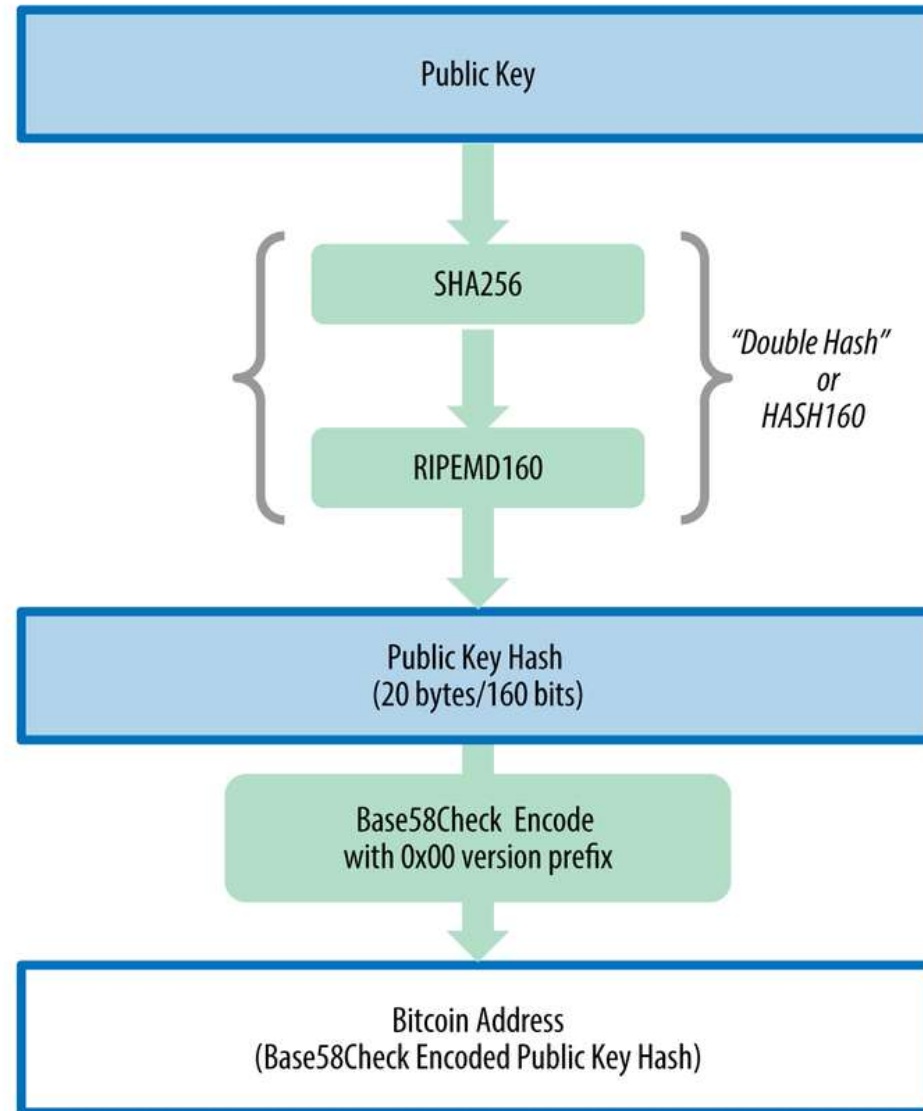
- The bitcoin address appears most commonly in a transaction as the “recipient” of the funds.
- The bitcoin address is derived from the public key through the use of one-way cryptographic hashing.
- A “hashing algorithm” or simply “hash algorithm” is a one-way function that produces a “hash” of an arbitrary-sized input.

- The algorithms used to make a bitcoin address from a public key are the Secure Hash Algorithm (SHA) and the RACE Integrity Primitives Evaluation Message Digest (RIPEMD), specifically SHA256 and RIPEMD160.
- Starting with the public key K , we compute the SHA256 hash and then compute the RIPEMD160 hash of the result, producing a 160-bit (20-byte) number:

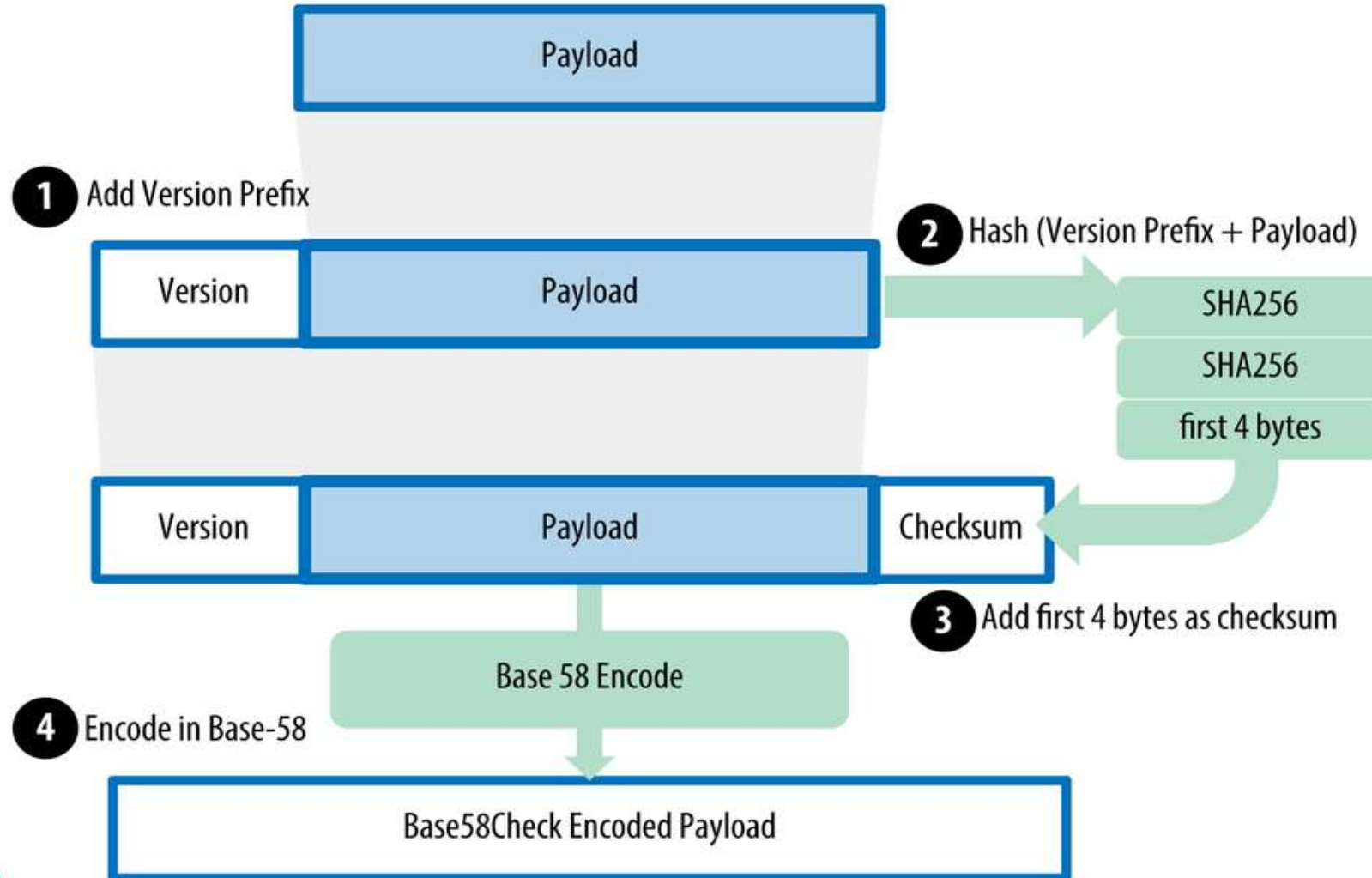
$$A = RIPEMD160(SHA256(K))$$

- Where k is the public key and A is the bitcoin address.

Public Key to Bitcoin Address



Base58Check Encoding



- To add extra security against typos or transcription errors, Base58Check is a Base58 encoding format, frequently used in bitcoin, which has a built-in error-checking code.
- The checksum is an additional four bytes added to the end of the data that is being encoded. The checksum is derived from the hash of the encoded data and can therefore be used to detect and prevent transcription and typing errors.
- When presented with a Base58Check code, the decoding software will calculate the checksum of the data and compare it to the checksum included in the code. If the two do not match, that indicates that an error has been introduced and the Base58Check data is invalid.

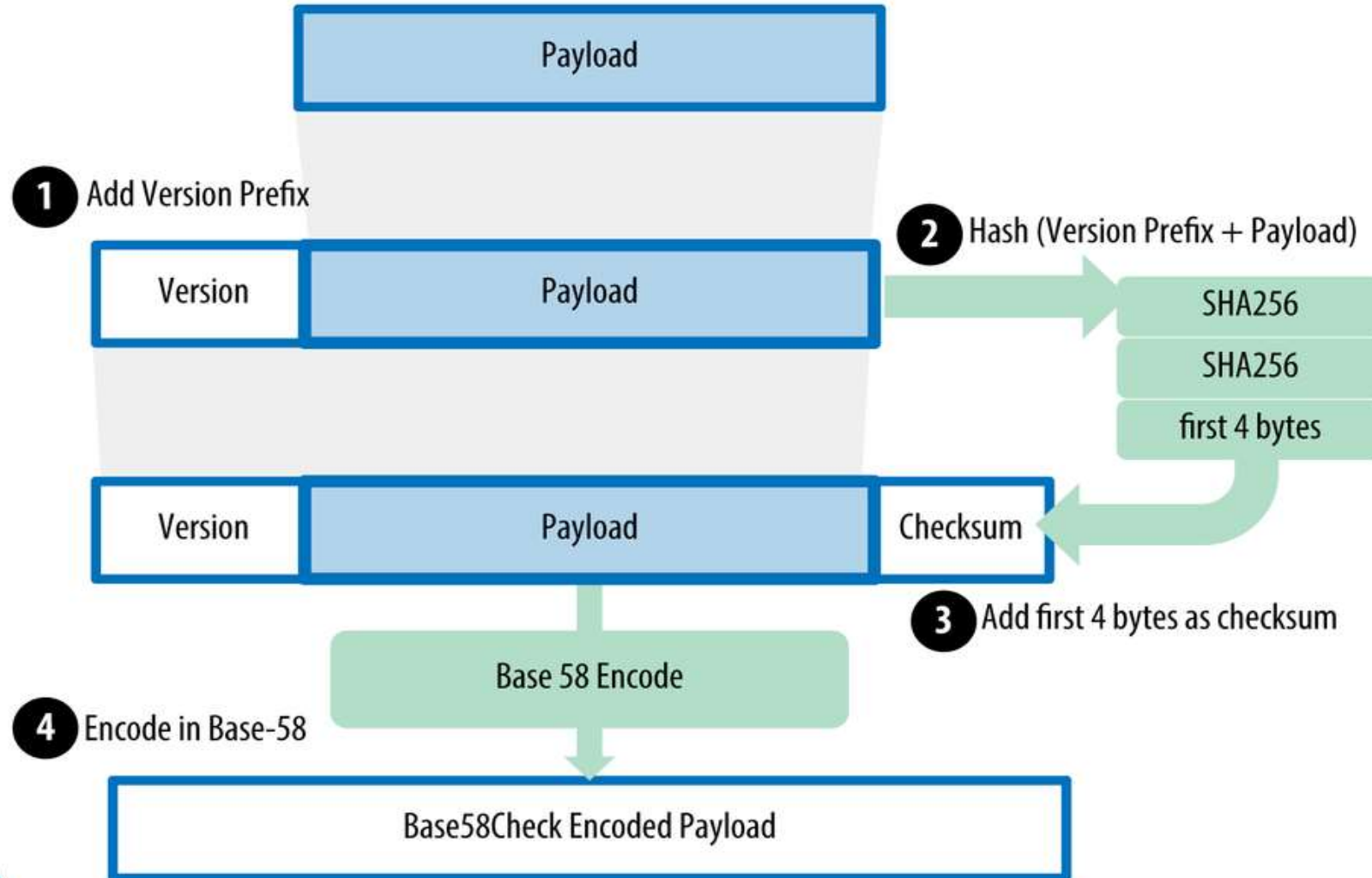
Type	Version prefix (hex)	Base58 result prefix
Bitcoin Address	0x00	1
Pay-to-Script-Hash Address	0x05	3
Bitcoin Testnet Address	0x6F	m or n
Private Key WIF	0x80	5, K or L
BIP38 Encrypted Private Key	0x0142	6P
BIP32 Extended Public Key	0x0488B21E	xpub

- Next, we compute the “double-SHA” checksum, meaning we apply the SHA256 hash-algorithm twice on the previous result (prefix and data):

checksum = SHA256(SHA256(prefix+data))

- From the resulting 32-byte hash (hash-of-a-hash), we take only the first four bytes. These four bytes serve as the error-checking code, or checksum. The checksum is concatenated (appended) to the end.
- The result is composed of three items: a prefix, the data, and a checksum.

Base58Check Encoding



KEY FORMATS

- Private Key Formats
- Public Key Formats

PRIVATE KEY ENCODING FORMATS

Type	Prefix	Description
Hex	None	64 hexadecimal digits 1e99423a4ed27608a15a2616a2b0e9e52ced330ac530ed cc32c8ffc6a526aedd
WIF	5	Base58Check encoding: Base58 with version prefix of 128 and 32-bit checksum 5J3mBbAH58CpQ3Y5RNJpUKPE62SQ5tfcvU2JpbnkeyhfsY B1Jcn
WIF-compressed	K or L	As above, with added suffix 0x01 before encoding KxFC1jmwwCoACiCAWZ3eXa96mBM6tb3TYzGmf6YwgdG WZgawvrtJ

Public Key Formats

- Compressed Public Keys: To reduce the size of transaction and conserve disk space on nodes that the bitcoin blockchain database. The transactions include the public key, required to validate the owner's credentials and spend the bitcoin.
- Each public key requires 520 bits (prefix $\{x, y\}$), which when multiplied by several hundred transactions per block, or tens of thousands of transactions per day, adds a significant amount of data to the blockchain.
- Uncompressed Public Keys

Wallets

- Wallets are containers for private keys, usually implemented as structured files or simple databases.

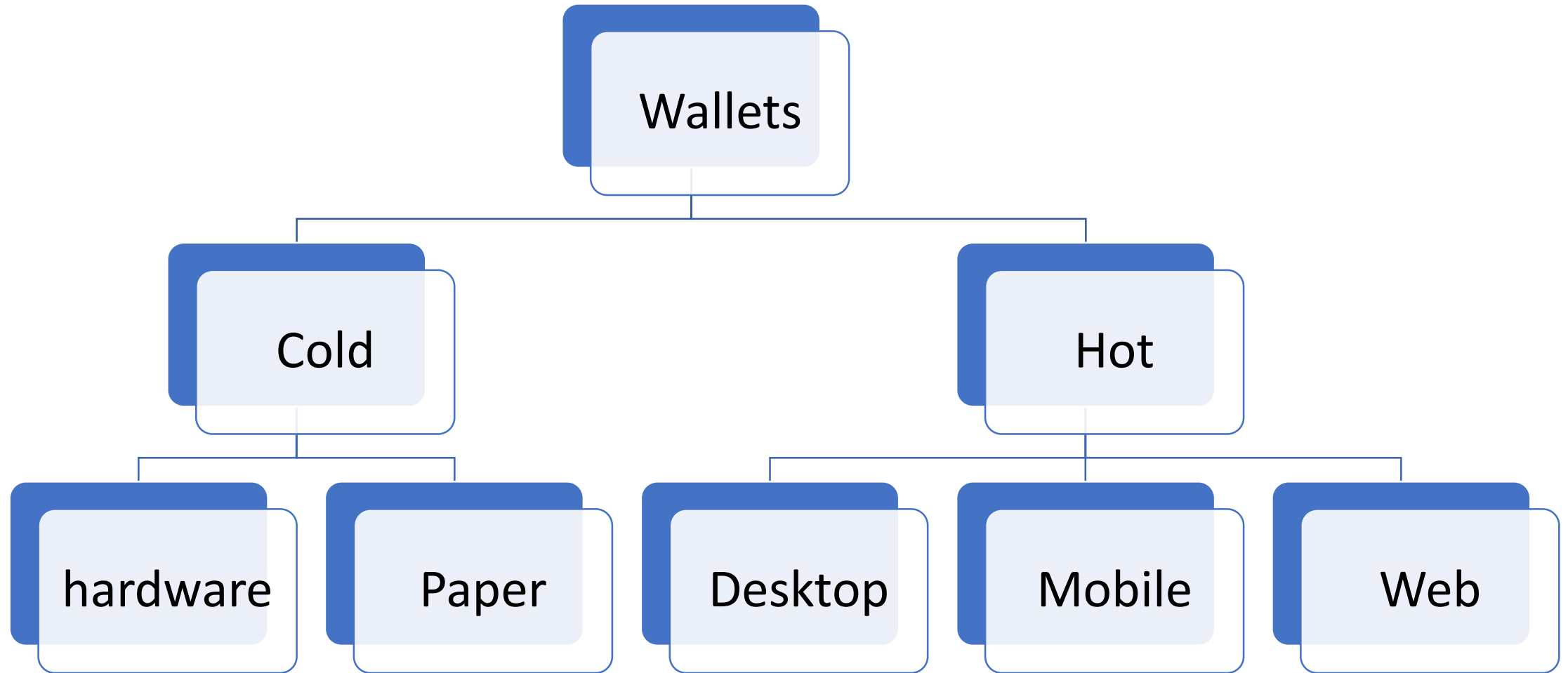
Crypto Wallet

- A crypto wallet is used to interact with a Blockchain network. Crypto wallets don't store the currency but act as a tool of interaction with Blockchain, i.e., generating the necessary information to receive and send money via Blockchain transactions.

FEATURES OF A CRYPTO WALLET

- A crypto wallet holds a public key for receiving funds and a private key for access and transactions.
- It allows users to send and receive cryptocurrencies by signing transactions with their private keys.
- Wallets can be categorized based on their internet connectivity (hot or cold) and the method of storage (hardware, software, or paper).
- Some wallets offer features like two-factor authentication (2FA), multi-signature support, and recovery options to enhance security and usability.

TYPES OF WALLETS



HOT AND COLD WALLETS

- Internet connectivity defines a wallet in terms of hot or cold.
- Hot wallets are connected to the Internet and thus are less secure and pose more risks but are user-friendly. Hot wallets are more likely to be used for daily transactions. Hot wallets are easy to set up, and the funds are quickly accessible.
- Cold wallets are stored offline and don't require internet connectivity. Cold wallets for more long-term holdings. Cold wallets are hack resistant, and thus the cold storage is suitable for HODLers.

HARDWARE WALLET

- Hardware wallets are hardware devices that individually handle public addresses and keys. It looks like a USB with an OLED screen and side buttons.
- It is a battery-less device and can be connected to a PC and accessed by native desktop apps. It costs up to 70-150 dollars, but it is worth it. They have received a mixed response.
- They are more secure than hot wallets and user-friendlier than paper wallets but less than web and desktop wallets. They are available in different forms and offer reasonable amounts of control.
- They are difficult for beginners to use when the investment is significant. The Most popular hardware wallets are Ledger Nano S and Trezor.

PAPER WALLETS

- It is a physically printed QR coded form wallet. Some wallets allow downloading the code to generate new addresses offline.
- They are not prone to hacks, but the number of flaws has made them dangerous. A major flaw is not being able to send partial funds. Thus, it can't be reused.
- They used to be very popular for cold storage, but not after hardware wallets came onto the scene.

DESKTOP WALLET

- These are installable software packs available for operating systems and are becoming serious with time.
- Anti-virus is required because a system connected to the Internet poses fundamental security issues. Instead of keeping cryptos on an exchange, desktop wallets for bitcoins should be used.
- They are the third most secure way to store cryptocurrencies and the best method for cold storage in a completely clean system. They are easy to use, give privacy and anonymity, and involve no third party.
- Regular backing up of the computer is needed. Popular desktop wallets are Exodus, Bitcoin core, Electrum, etc.

MOBILE WALLETS

- Mobile wallets are just like desktop wallets made for smartphones. They are quite convenient as it uses QR codes for transactions. They are suitable for daily operations but are vulnerable to malware infection.
- Encryption of mobile wallets is necessary. They are practical and can be used on the go but open to viruses.
- Some mobile wallets are Coinomi and Mycelium.

WEB WALLETS

- As the name suggests, these wallets are accessed by internet browsers. The private keys are held in some web wallets and are prone to DDOS attacks. They can be hosted or non-hosted. Non-hosted is preferred as funds are always in control. They are the least secure wallets. They are not the same as hot wallets. They are ideal for small investments and allow quick transactions. Some of these are MetaMask and Coinbase.

CUSTODIAL WALLETS

- Custodial wallets are provided by third-party service providers, such as cryptocurrency exchanges or online wallet platforms. These providers take care of managing the private keys and security of the cryptocurrencies on the user behalf.
- Custodial wallets are often designed to be user-friendly, making them a suitable choice for beginners or those who prefer a more straightforward approach to managing their digital assets. They are generally easier to set up and use.
- Custodial wallet providers offer account recovery options in case the user forgets the password or lose access to the account. This can be convenient, but it also means that the service provider has a level of control over the user assets.
- The main risk with custodial wallets is that the service provider can be a target for hackers. If the provider's security is breached, the funds could be at risk. Additionally, the service provider may have terms and conditions that could affect the access to the funds.
- Some examples of custodial wallets include wallets provided by popular cryptocurrency exchanges like Coinbase, Binance, and Kraken.

NON-CUSTODIAL WALLETS

- Non-custodial wallets provides complete control over the private keys and, consequently, the cryptocurrencies. The user is responsible for the security and management of the assets.
- Non-custodial wallets prioritize security and privacy. Since user hold the private keys, it's less likely that the funds will be compromised through a breach of a third-party service. This makes them a preferred choice for individuals who value security.
- Non-custodial wallets are known for offering a higher level of privacy and anonymity. Financial transactions and holdings are not subject to the data collection practices of third-party providers.
- While non-custodial wallets offer more control, they also require users to take responsibility for their private keys and ensure they are kept secure. Losing access to the private keys can result in the loss of funds, with no recovery options.
- Examples of non-custodial wallets include hardware wallets like Ledger Nano S and Trezor, as well as software wallets like Electrum, MyEtherWallet, and Trust Wallet.

WORKING PRINCIPLE OF CRYPTO WALLETS

Step	Description
1. Acquiring a Wallet	Receive a cryptocurrency wallet when digital assets are purchased. Initial wallet is typically a hot wallet.
2. Understanding Wallet Types	Differentiate between hot wallets (online) and cold wallets (offline) based on security preferences.
3. Transferring to Cold Wallet (Optional)	Enhance security by transferring cryptocurrency from a hot wallet to a cold wallet (recommended for long-term storage).
4. Understanding Cryptocurrency Storage	Realize that cryptocurrencies exist as data on a Blockchain, a decentralized ledger.
5. Key Components	A wallet holds a public key (for receiving) and a private key (for access and transactions).
6. Accessing and Transacting	Use the private key, similar to a digital signature, to access and conduct cryptocurrency transactions.
7. Sending Cryptocurrency	When sending, enter the recipient's address, specify the amount, sign the transaction with the private key, add a transaction fee, and confirm the send.
8. Receiving Cryptocurrency	To receive, the sender uses your cryptocurrency address as the destination, and you accept the payment, which is confirmed on the Blockchain.

Crypto wallet Real time issues

- The LastPass Incident
- BitForge Vulnerabilities
- The MetaMask Incident
- Phishing Attacks
- Vulnerable Key Management Systems

PROTECTION OF CRYPTO WALLETS

Security	Description
Hardware Wallets	Use physical devices (e.g., Ledger, Trezor) to keep keys offline for maximum security.
Two-Factor Authentication (2FA)	Enable 2FA for extra login protection by requiring a mobile-generated code.
Multi-Signature Wallets	Employ multi-signature wallets for shared accounts, reducing single points of failure.
Secure Passwords	Use strong, unique passwords and consider password managers.
Seed Phrase Backup	Safely store offline as a recovery key, split it for added security.
Cold Storage	Keep assets offline in ‘cold wallets’ like paper wallets.
VPN for Extra Shielding	Use a VPN for encryption and privacy while accessing your wallet.
Multiple Wallets	Diversify assets across various wallets for risk reduction.
Trusted Sources	Buy hardware wallets from reputable vendors or official sources.
Secure Environment	Use secure, malware-free devices and avoid risky actions.

THE TRANSACTION LIFE CYCLE

1. User/sender sends a transaction using wallet software or some other interface.
2. The wallet software signs the transaction using the sender's private key.
3. The transaction is broadcasted to the Bitcoin network using a flooding algorithm.
4. Mining nodes (miners) who are listening for the transactions verify and include this transaction in the next block to be mined. Just before the transaction are placed in the block they are placed in a special memory buffer called transaction pool.
5. Mining starts, which is a process by which the blockchain is secured and new coins are generated as a reward for the miners who spend appropriate computational resources.
6. Once a miner solves the PoW problem it broadcasts the newly mined block to the network.
7. The nodes verify the block and propagate the block further, and confirmations start to generate.
8. Finally, the confirmations start to appear in the receiver's wallet and after approximately three confirmations, the transaction is considered finalized and confirmed. However, three to six is just a recommended number; the transaction can be considered final even after the first confirmation. The key idea behind waiting for six confirmations is that the probability of double spending is virtually eliminated after three confirmations.

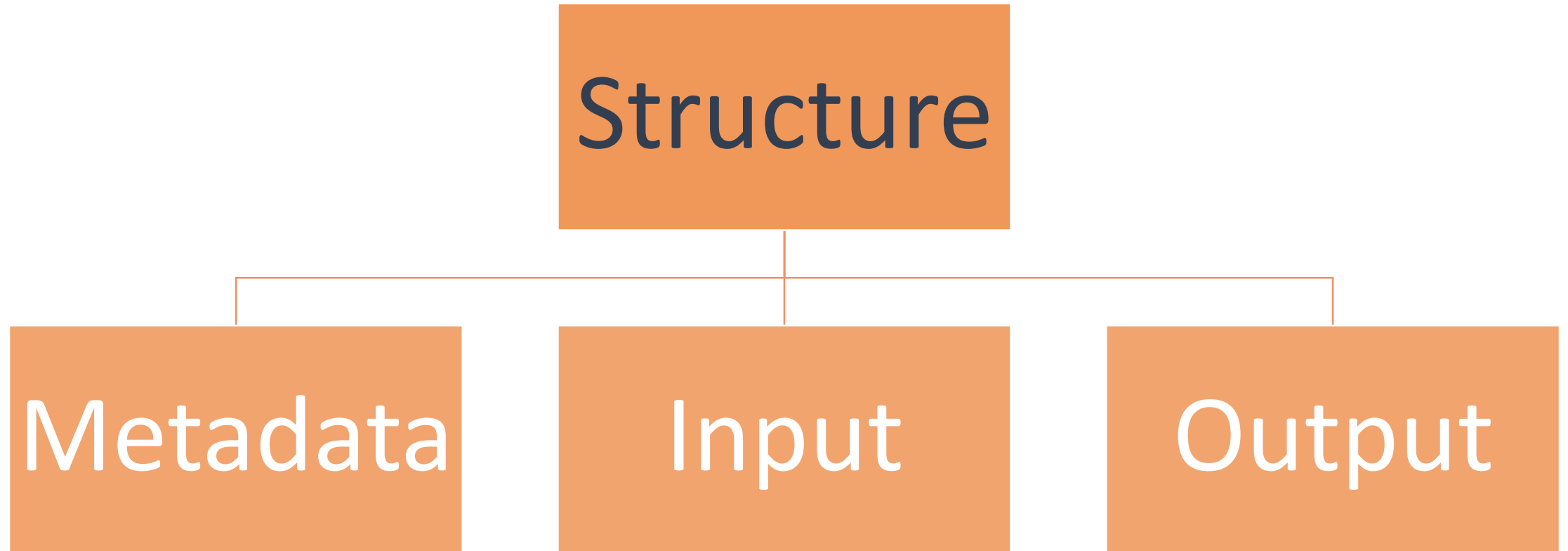
TRANSACTION FEES

- A Blockchain transaction fee is a small amount that is charged to the users who send transactions to the Blockchain network. These fees help in the mining process and reward the miners for providing their computing power for the operations.
- The fee amount can vary based on the congestion of the network and the size of the transaction.

$$\text{FEE} = \text{SUM}(\text{INPUT}) - \text{SUM}(\text{OUTPUT})$$

- The transaction fee is a nominal amount that the user has to pay to the miners to have their transaction verified and added to the Blockchain ledger.
- Each Blockchain network has a maximum limit on the transaction fee users can set for their transactions.
- The transaction fee can be set at a nominal amount, or it can be set at zero.

Transaction Structure



Metadata

- Metadata: This part of the transaction contains some values such as the size of the transaction, the number of inputs and outputs, the hash of the transaction, and a lock_time field.
- Every transaction has a prefix specifying the version number.

INPUTS

Each input spends a previous output. Each output is considered as Unspent Transaction Output (UTXO) until an input consumes it. UTXO is an unspent transaction output that can be spent as an input to a new transaction.

Field	Size	Description
Transaction hash	32 bytes	This is the hash of the previous transaction with UTXO.
Output index	4 bytes	This is the previous transactions output index, that is, UTXO to be spent.
Script length	1-9 bytes	This is the size of the unlocking script.
Unlocking script	Variable	Input script (<code>ScriptSig</code>) which satisfies the requirements of the locking script.
Sequence number	4 bytes	Usually disabled or contains lock time. Disabled is represented by <code>'0xFFFFFFFF'</code> .

OUTPUTS

Field	Size	Description
Value	8 bytes	Total number in positive integers of Satoshis to be transferred
Script size	1-9 bytes	Size of the locking script
Locking script	Variable	Output script (<code>ScriptPubKey</code>)

TRANSACTION DATA STRUCTURE

FIELD	SIZE	DESCRIPTION
Version number	4 bytes	Used to specify rules to be used by the miners and nodes for transaction processing.
Input counter	1-9 bytes	The number (positive integer) of inputs included in the transaction.
List of inputs	Variable	Each input is composed of several fields, including Previous Tx hash, Previous Txout-index, Txin-script length, Txin-script, and optional sequence number.
Output counter	1-9 bytes	A positive integer representing the number of outputs
List of outputs	Variable	Outputs included in the transaction.
Lock time	4 bytes	This field defines the earliest time when a transaction becomes valid. It is either a Unix timestamp or block height

PURPOSE OF TRANSACTION FEES

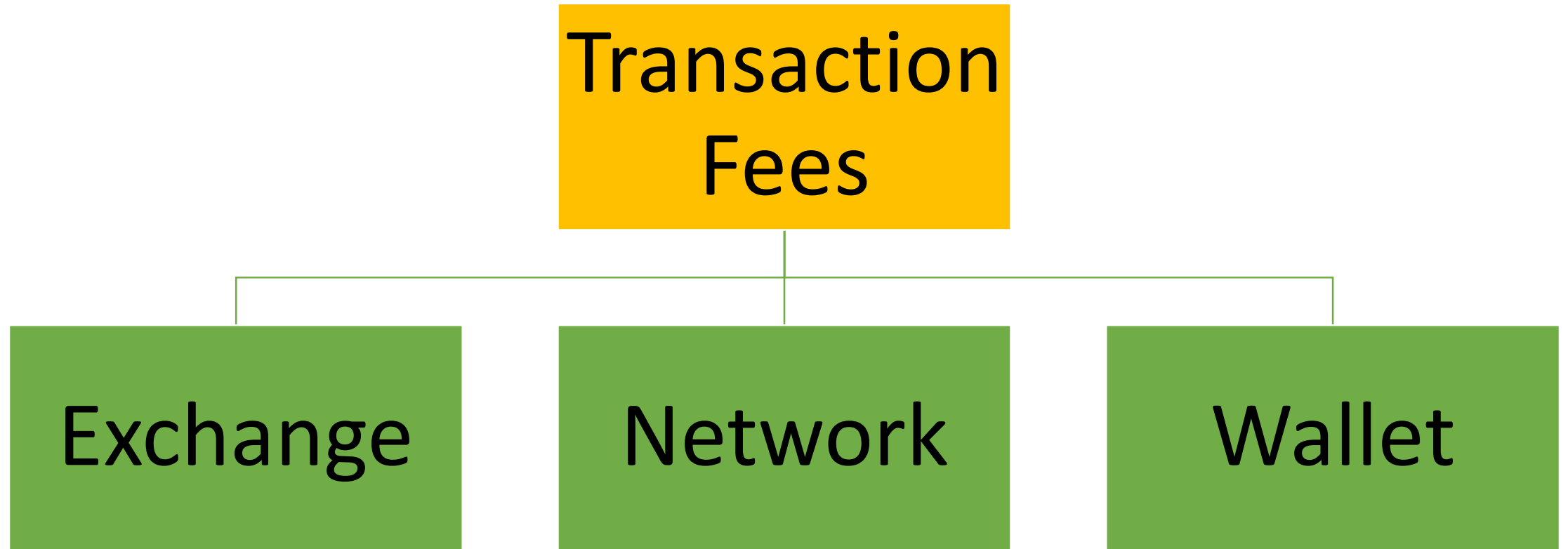
- **Reward**: To reward the miners for **validating** the transactions.. When users send a transaction to the Blockchain network, it is **verified** by the miners, who use their computing power to unlock new blocks in the Blockchain network.
- **Adding to block**: The miners also verify the transactions in the Blockchain and add them to the ledger.
- **Avoid Spam**: Transaction fees also help in avoiding spam transactions sent to the Blockchain network.
- **Faster**: When a transaction is send to the network and if it has to be processed fast then transaction fees is to be paid. Higher the fee faster the process.

- In this process, the miners receive the transaction fees in the form of the native cryptocurrencies of the Blockchain network.
- If there are no transaction fees, the miners will lose their incentive to validate the transactions because they won't get anything in return.
- When the transaction volume is high, it might take a few minutes or even hours to confirm the transactions.
- Most Blockchain networks have transaction limits to prevent malicious activities and denial-of-service (DoS) attacks. This means that if you send funds larger than the defined transaction limit, your transaction will be rejected.
- However, genuine transactions will also be rejected if the transaction limit is too low. This is where the transaction fee can significantly impact the transaction time.
- If a fee is set too high, it becomes costly to send funds. This can slow down the transaction process as the users will have to wait for their funds due to high fees.

Transaction fees calculation

- The transaction fees are paid in terms of cryptocurrencies.
- Determined in large part by due to the differing levels of complexity, data size, and urgency.
- The transaction fee is calculated based on the mining difficulty and mining reward.
- Mining Difficulty: number of miners who are mining on the Blockchain network. The mining difficulty of the Blockchain network is usually high, and it decreases when the mining difficulty of the Blockchain network is low.
- Mining Reward: The mining reward is related to the number of rewards the miners will receive for unlocking the next block in the Blockchain network.

Types of Transaction Fees



Exchange Fees

- Exchanges are websites and apps where users can buy and sell cryptocurrencies. These exchanges are centralized systems that are controlled by a single entity. Although these exchanges are a convenient way to trade and sell digital tokens, a fees has to be paid to the exchange.
- The transaction fee can be paid through the exchange's native token or other popular cryptocurrencies like Bitcoin and Ether.

Network Fees

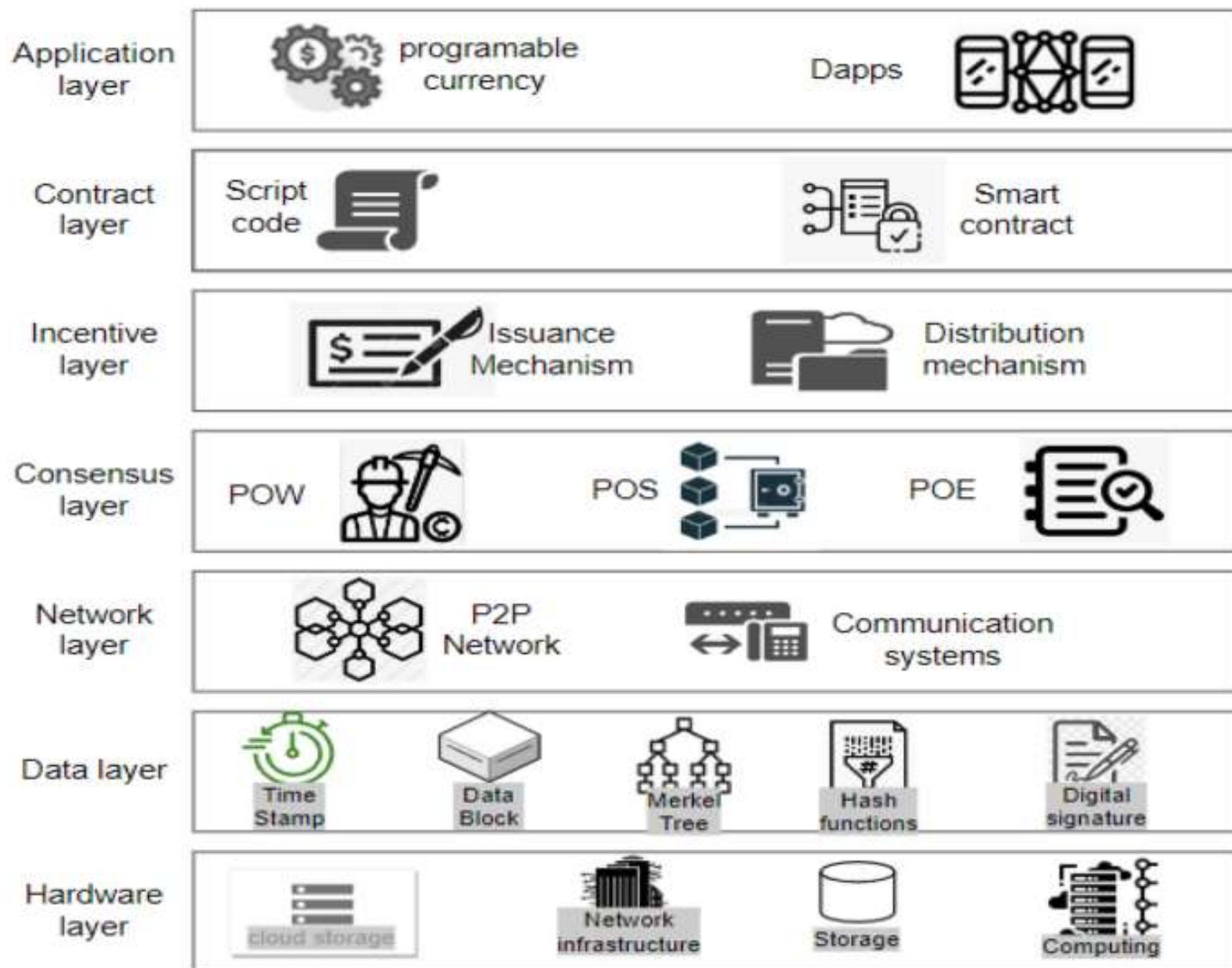
- Network fees are the transaction fees charged by the underlying Blockchain network. These fees are charged for adding a new transaction to the Blockchain network.
- Network fees ensure the efficient and secure functioning of the system, as they deter spam transactions and malicious activities, while encouraging miners/validators to prioritize transactions with higher fees.
- Blockchain networks might impose a “gas limit” for a specific transaction. This means that the transaction will only be confirmed if the sender pays the appropriate network fee. These network fees can be very high during periods of high network usage and can significantly impact the transaction time.

Wallet fees

- Wallet fees are the transaction fees charged by the wallet service provider. A wallet is an online service where one can store the cryptocurrencies. These wallets charge a nominal fee on a per-transaction basis.

LAYERS OF BLOCKCHAIN

Application and Presentation Layer Smart Contract, Chaincode, DApps, UI
Consensus Layer Pow, Pos, DPos, PoET, PBFT
Network Layer Peer-to-Peer
Data Layer Digital Signature, Hash, Merkel Tree, Transactions
Hardware/Infrastructure Layer Virtual Machine, Containers, Mining Rig



HARDWARE/INFRASTRUCTURE LAYER

- This layer consists of the physical components that support the blockchain network, such as computers and servers.
- A node is a computer or network of computers that decrypts transactions, and a blockchain is the sum of all nodes. These nodes are spread across many geographical areas. They nodes are linked to one another as peer-to-peer using internet infrastructures.
- The computer network quantifies both the monetary and non-monetary transactions, verifies these transactions, and saves them in a common /mutual ledger shared by all the nodes that are participating.
- Data collected is stored in local nodes in the on-chain approach and remotely in the off-chain approach.

DATA LAYER

- The transaction information recorded on a block (the basic unit of a blockchain) includes information about the sent crypto, the public key of the recipient, and the private key of the sender.
- Metadata of the chain is usually stored in the block header, which is the Merkle tree root hash, a hash of the previous block, the current version of the block, and a time stamp. Whereas the block body usually possesses a transaction and transaction counter. After the data is added to the chain it cannot be mutated or modified.
- Hashing and Merkle tree are used in this layer for maintaining confidentiality and security

NETWORK LAYER

- This layer handles the communication between blockchain nodes. It connects nodes, discovers nodes, propagates transactions, and distributes data throughout the network.
- Since blockchain is an open system, each node must be aware of the transactions being validated by other nodes. The network layer facilitates this communication.
- To maintain the global state of the blockchain it is necessary to take care of the state propagation, this is also taken care of at this layer.

CONSENSUS LAYER

- This layer guarantees that all nodes in the network concur on the validity of each transaction.
- Reliability is a key aspect of blockchain and the consensus layer is responsible for this in the blockchain network. To achieve it every participant is required to follow all the set of rules that are being enforced by the layer and these rules are called consensus.
- It uses a consensus mechanism, such as Proof of Work (PoW) or Proof of Stake (PoS), to validate and add transactions to the blockchain.
- The consensus layer verifies, administers, maintains, and does the management of block generation.
- The consensus layer also rewards the validator node and mining node based on the performance. It uses many consensus to ensure consistency, but the two most widely used are probabilistic and deterministic approaches.

INCENTIVE LAYER

- The role of the incentive layer is to maximize node participation in security verification conducted by the blockchain.
- It is achieved by giving some incentives to the participating nodes. If participation increases, the security will also increase.
- The role of the incentive layer is to maximize node participation in security verification conducted by the blockchain. It is achieved by giving some incentives to the participating nodes.

CONTRACT LAYER

- The contract layer is also known as the smart contract layer. It is quite similar yet different from an auto-executable piece of code.
- It comprises several algorithms, multiple scripts and contracts which makes blockchain more manageable and programmable.
- It reacts to messages received or sent, it can store, and transfer values and information.

APPLICATION LAYER

- The Application layer in the blockchain is the one on which apps are built.
- This run on top of the blockchain network. It allows developers to create new applications and services that leverage the security and transparency of the blockchain.
- These implementations may consist of anything, like wallets, social media Apps, browsers, Defi Apps, and NFT platforms, to name a few.
- While the UI/UX of the app is identical to that of any other standard application, the backend data storage of these applications is decentralized.